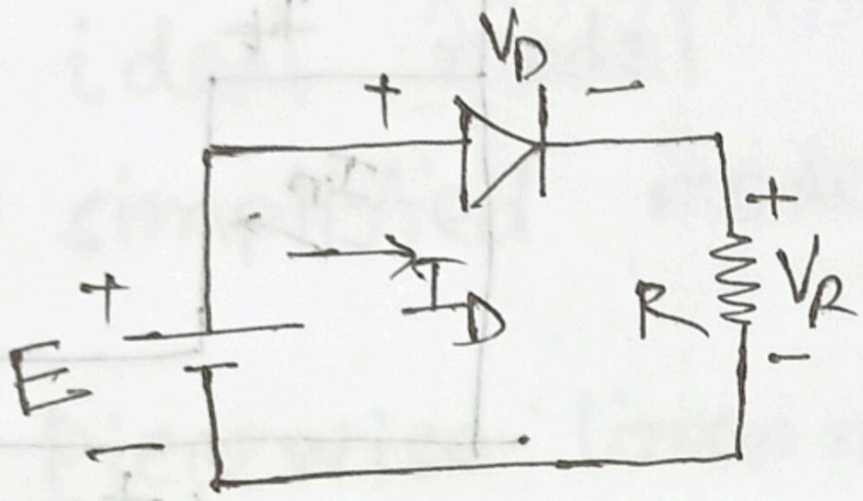
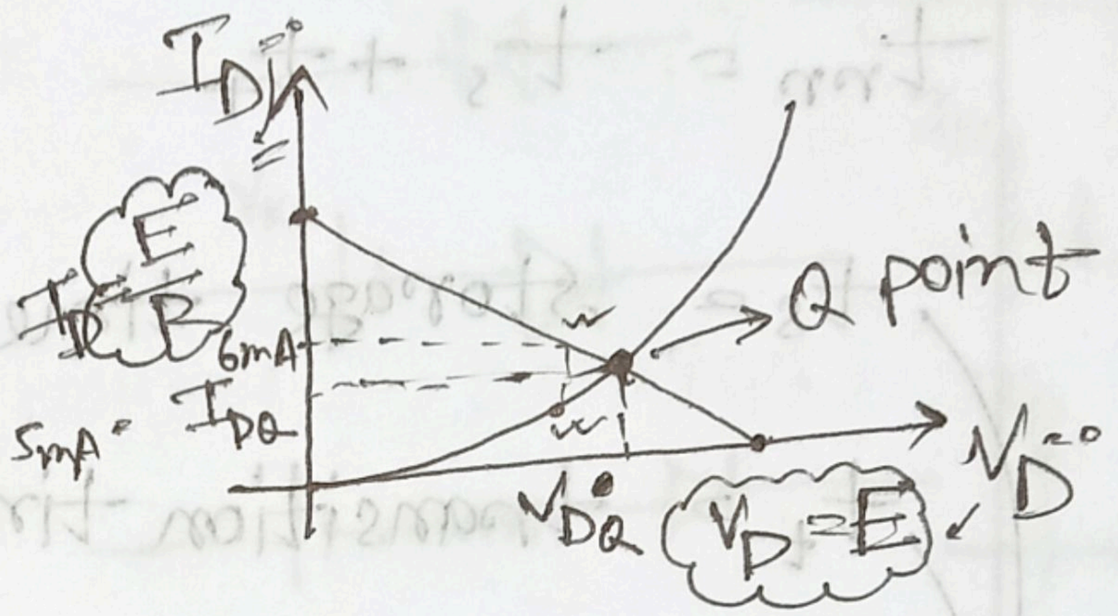
Chapter-02

Load line analysis:





$-E + V_D + I_D R = 0$

$\Rightarrow E = V_D + I_D R$

$\Rightarrow I_D = \frac{E}{R} - \frac{V_D}{R}$

load line analysis এর জন্য I_D এর মানের রেঞ্জ বের করতে হবে।

max এবং min বিন্দু।

I_D max.

$V_D = 0$

$\therefore I_D = \frac{E}{R}$ (when $V_D = 0$)

I_D min.

$I_D = 0$

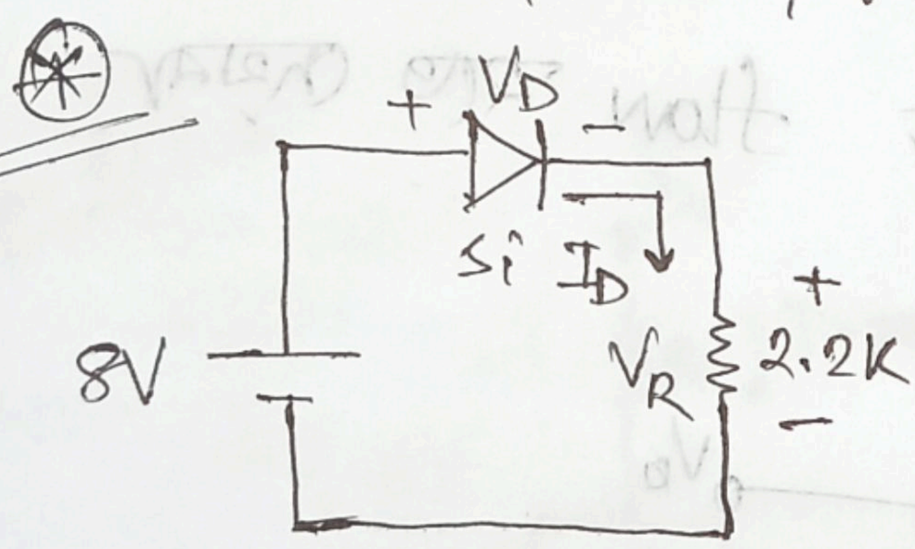
$E = V_D|_{I_D=0}$

load line analysis এর জন্য। কারণ, Q point বলে দেয় circuit এ কী পরিমান voltage (V_{DQ}) এবং কী পরিমান current (I_{DQ}) থাকবে।

Example - 2.1 & 2.2 →

Series Diode Configuration

$Si = 0.7V$



$E = V_D + V_R$

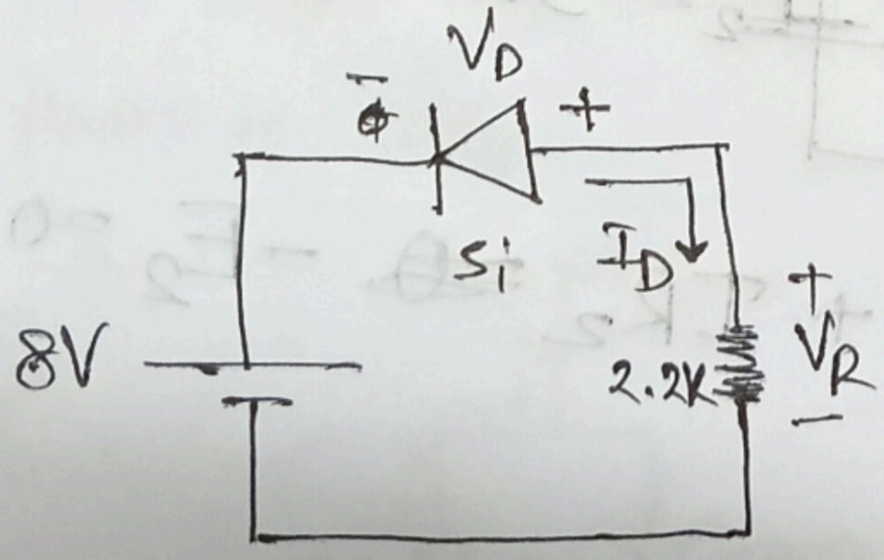
$V_D = 0.7V$

$V_R = 8 - 0.7$

$= 7.3V$

$I_D = I_R = \frac{7.3}{2.2K}$

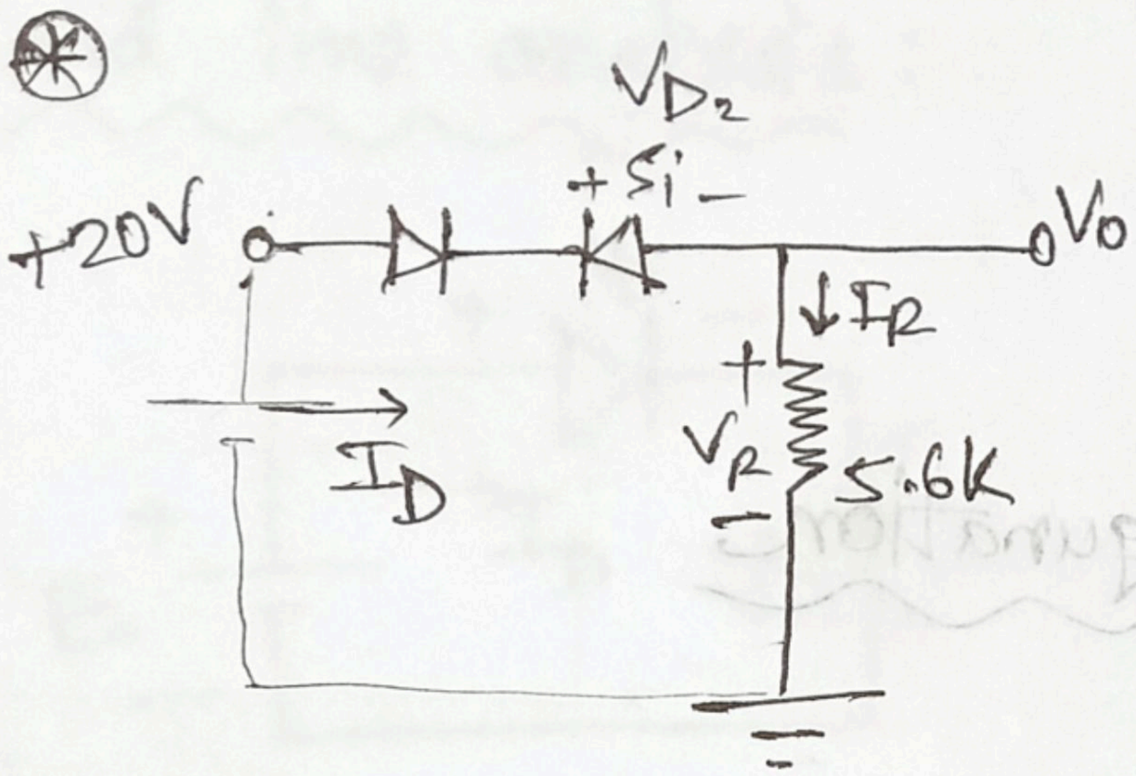
$= 3.32mA$



$I_D = 0$

$V_R = 0$

$V_D = 8V$

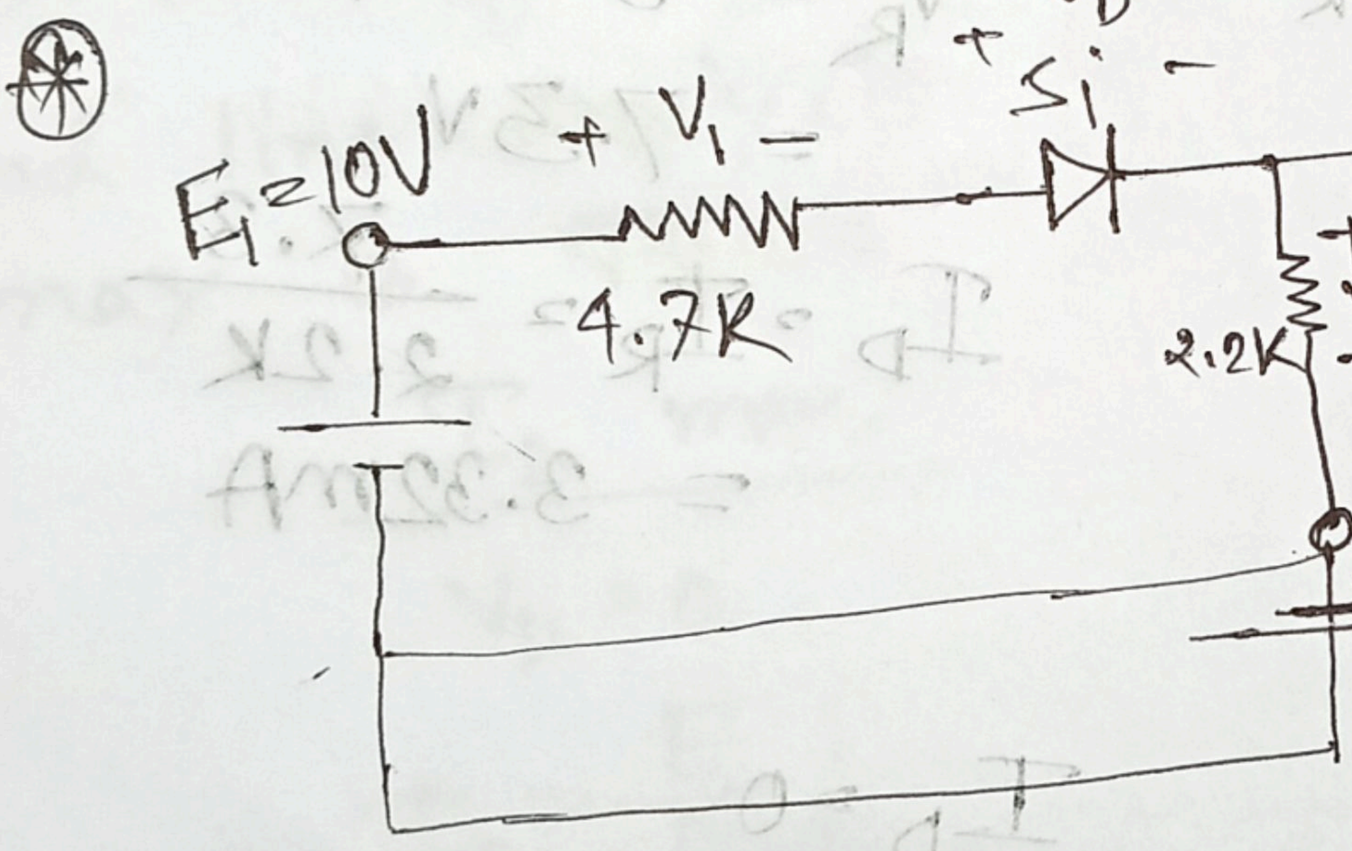


$I_D = 0$

$V_o = 0$

$V_{D_2} = 20V$

(K) কর্ম current flow হচ্ছে কোথায়



$$-E_1 + IR_1 + \cancel{0.7}\, V_D + IR_2 \cancel{-0} - E_2 = 0$$

$$\Rightarrow I = \frac{E_1 + E_2 - V_D}{R_1 + R_2}$$

$$= 2.07mA$$

$V_1 = IR_1 =$

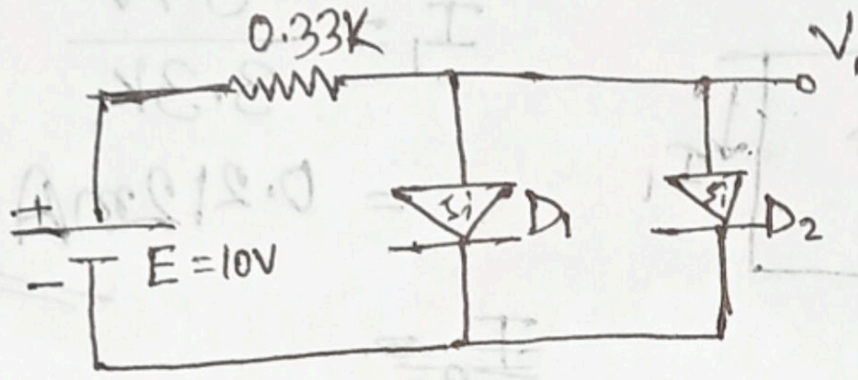
$V_2 = IR_2 =$

$V_o =$

Lec-4

26.10.26

Ex-2.10



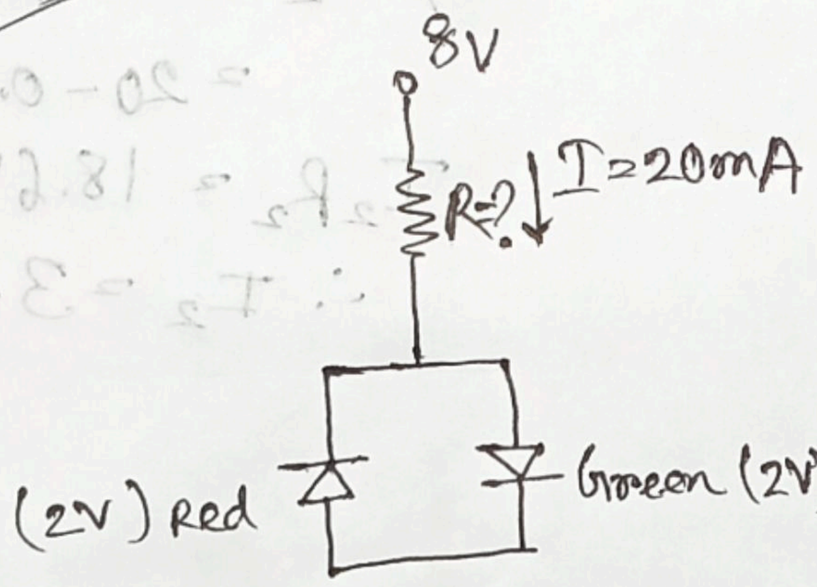
$V_o = 0.7V$

$I_1 = \frac{V_R}{R} = \frac{10-0.7}{0.33K}$

$= 28.18mA$

$I_{D_1} = I_{D_2} = \frac{I_1}{2} = 14.09mA$

Ex-2.11



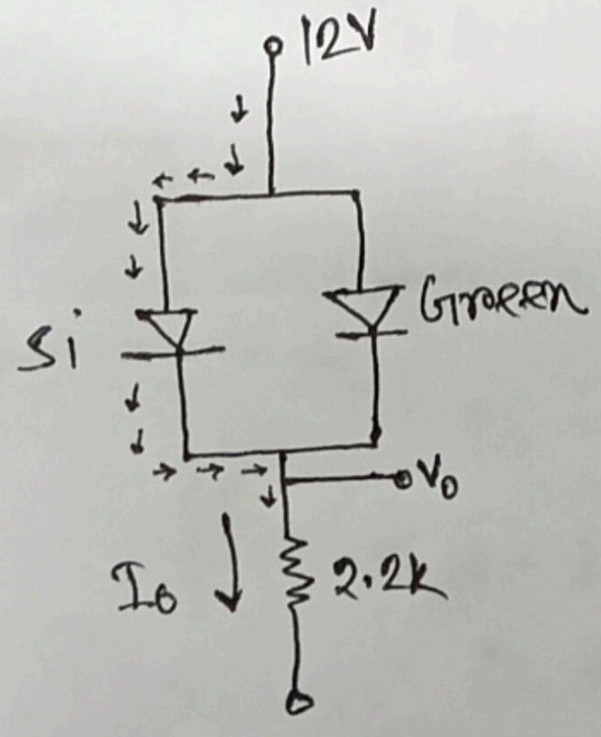
$I = \frac{8-2}{R}$

$\Rightarrow R = \frac{6}{20mA}$

$\therefore R = 0.3K\Omega$

Red Light জ্বলে কারেন্ট reverse bias.

Ex-2.12



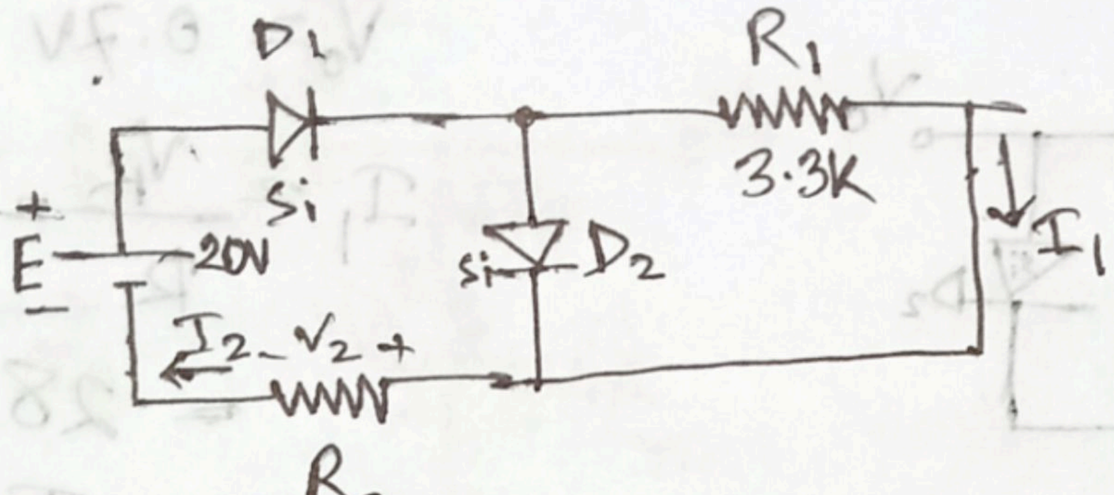
$V_o = 12 - 0.7$

$= 11.3V$

$I_o = \frac{11.3}{2.2K}$

$= 5.136mA$

Ex-2.13



$$I_1 = \frac{0.7V}{3.3K}$$

$$= 0.212mA$$

KVL,

$$-E + V_{D_1} + V_{D_2} + V_2 = 0$$

$$\Rightarrow V_2 = E - V_{D_1} - V_{D_2}$$

$$= 20 - 0.7 - 0.7$$

$$I_2 R_2 = 18.6V$$

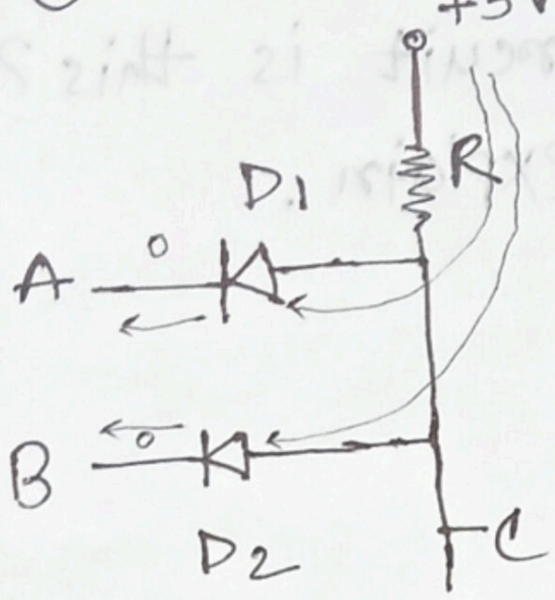
$$\therefore I_2 = 3.32mA$$

$$I_{D_2} = I_2 - I_1$$

$$= 3.32 - 0.212$$

$$= 3.11\ mA$$

AND Gate using Diode:



logic 1 → +5V
logic 0 → 0V

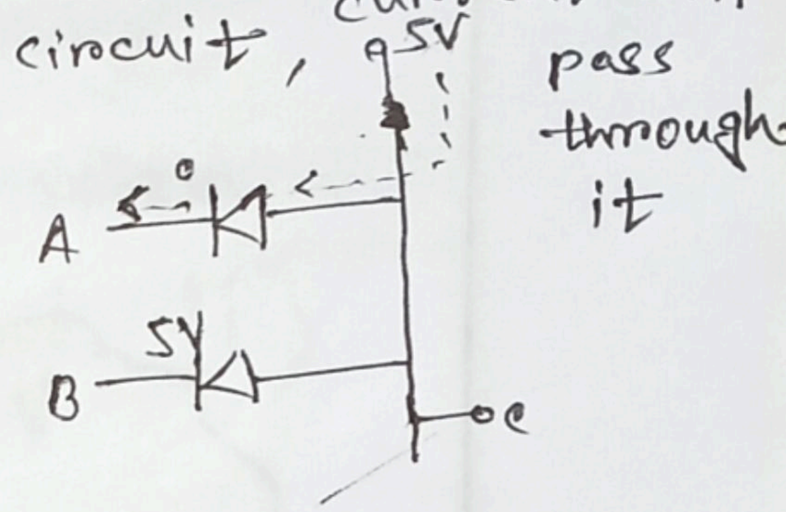
A=0, B=0; C=0

A=1, B=0; C=0

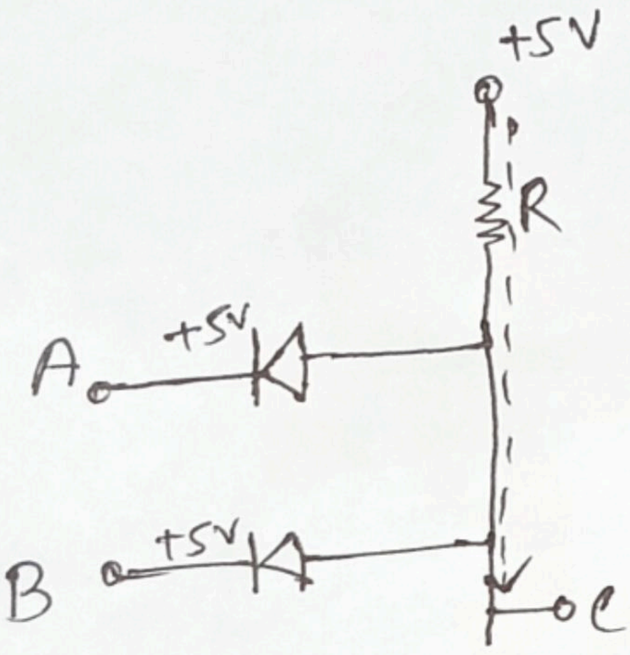
A=0, B=1; C=0

→ D1 = ON, D2 = OFF

Cause, A1 is short circuit, current will pass through it



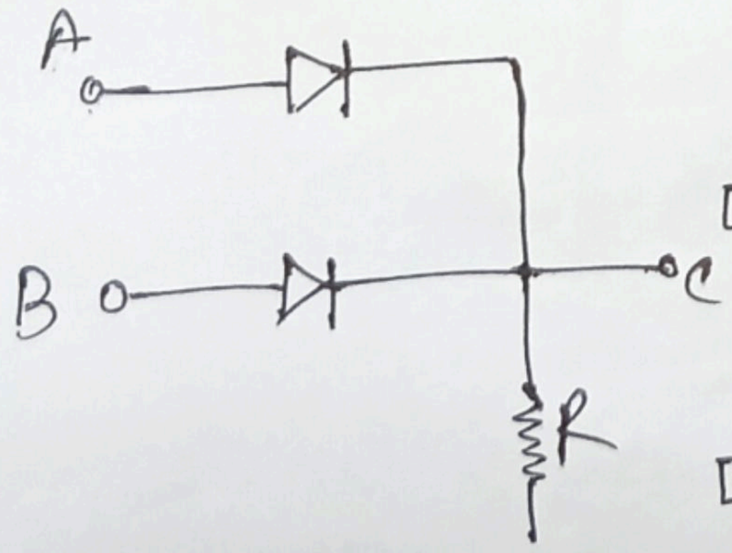
A	B	C
0	0	0
0	1	0
1	0	0
1	1	1



A=1, B=1; C=1

D1 = D2 = OFF

OR Gate using Diode:



A=0; B=0; C=0

D1 = OFF, D2 = OFF

A=1; B=0; C=1

D1 = ON; D2 = OFF

A=1; B=1; C=1

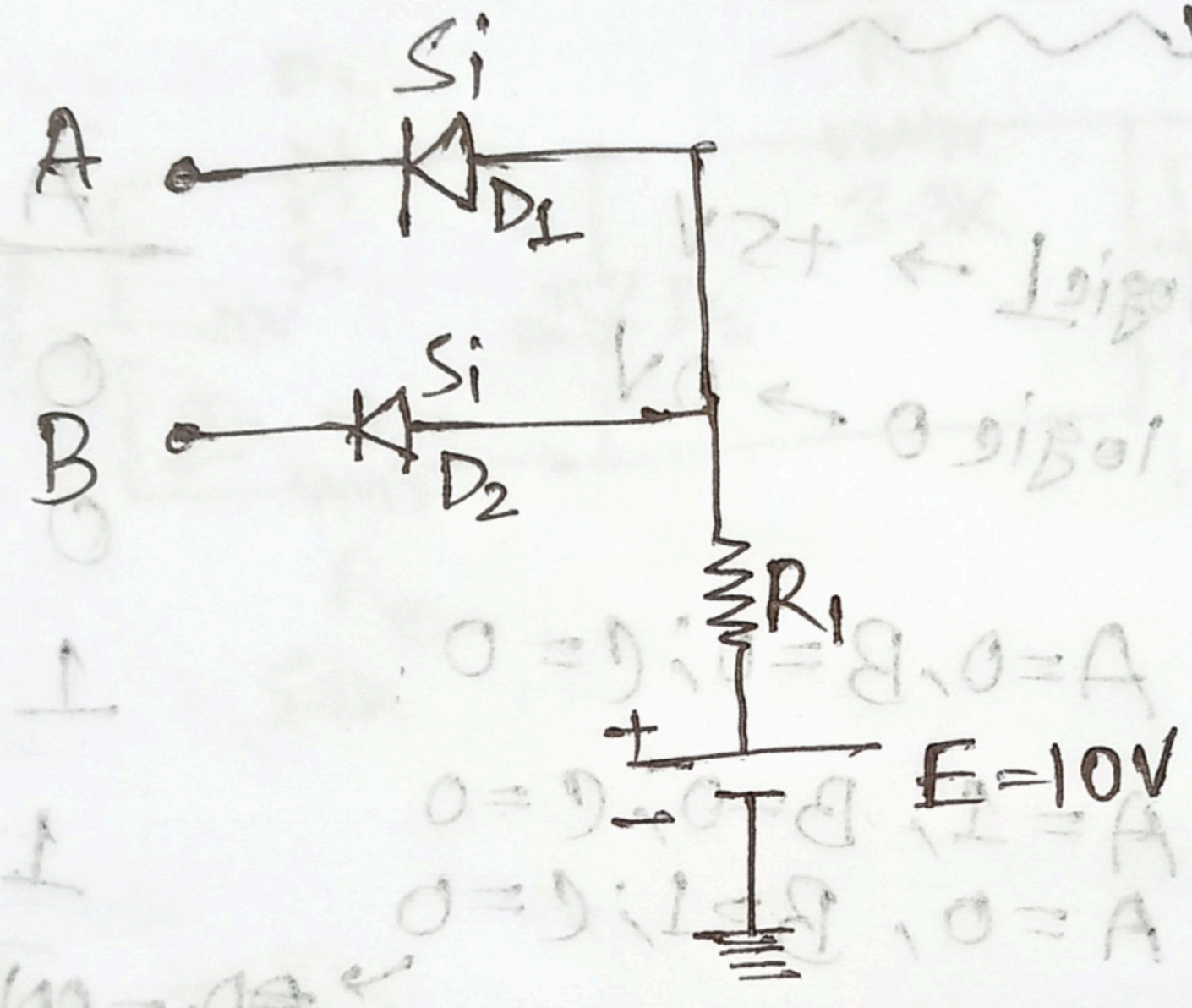
D1 = ON; D2 = ON

A	B	C
0	0	0
0	1	1
1	0	1
1	1	1

logic 0 = 0V

logic 1 = +5V

Task



Which type of gate/ circuit is this? explain.

→ And Gate.

28.10.26

Clipper Circuit

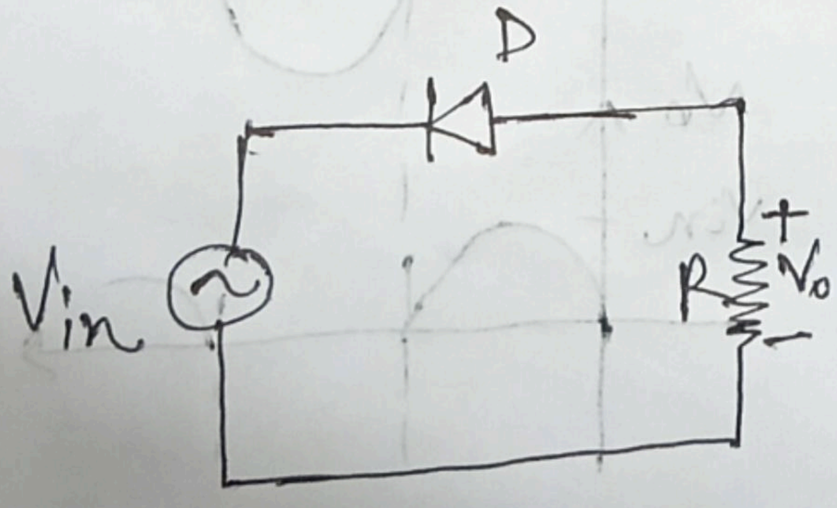
→ input এর কিছু portion কেটে দিবে।
→ Voltage limited বানিয়ে দিবে।

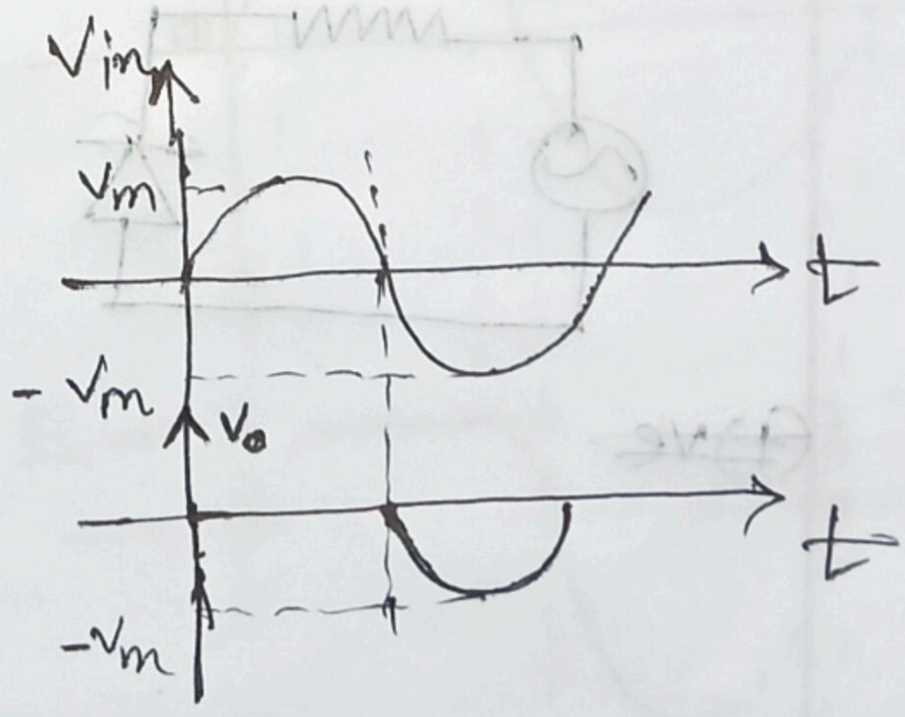
① Positive Clipper → Positive series, Positive shunt
② Negative " → series, shunt
③ Biased " → Positive bias, Negative bias
④ Combinational "

positive clipper

① Positive series clipper

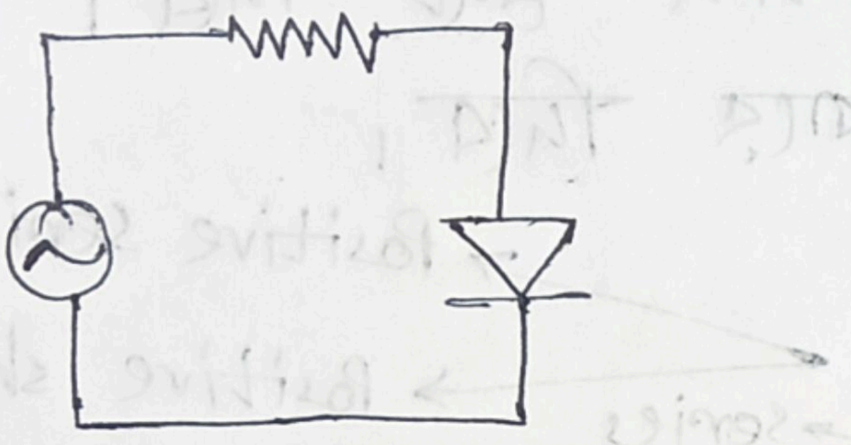
→ Current waveform কে limit করে।

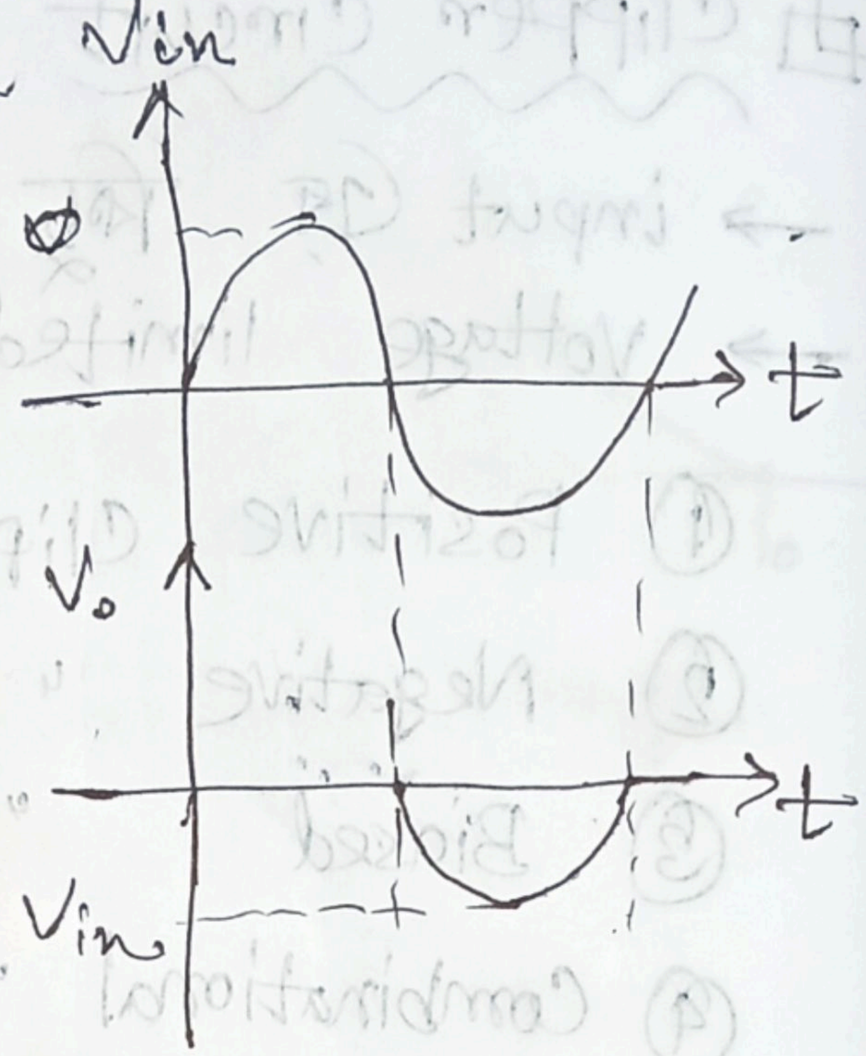




+ve half cycle →
D ⇒ off, $V_o = 0$

② Positive Shunt Clipper





(+ve) half,

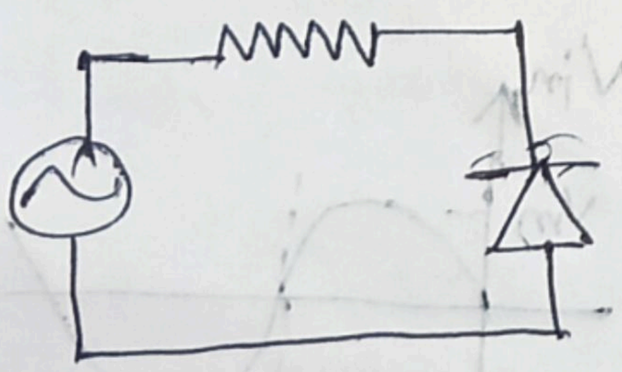
D = on; $V_o = 0$

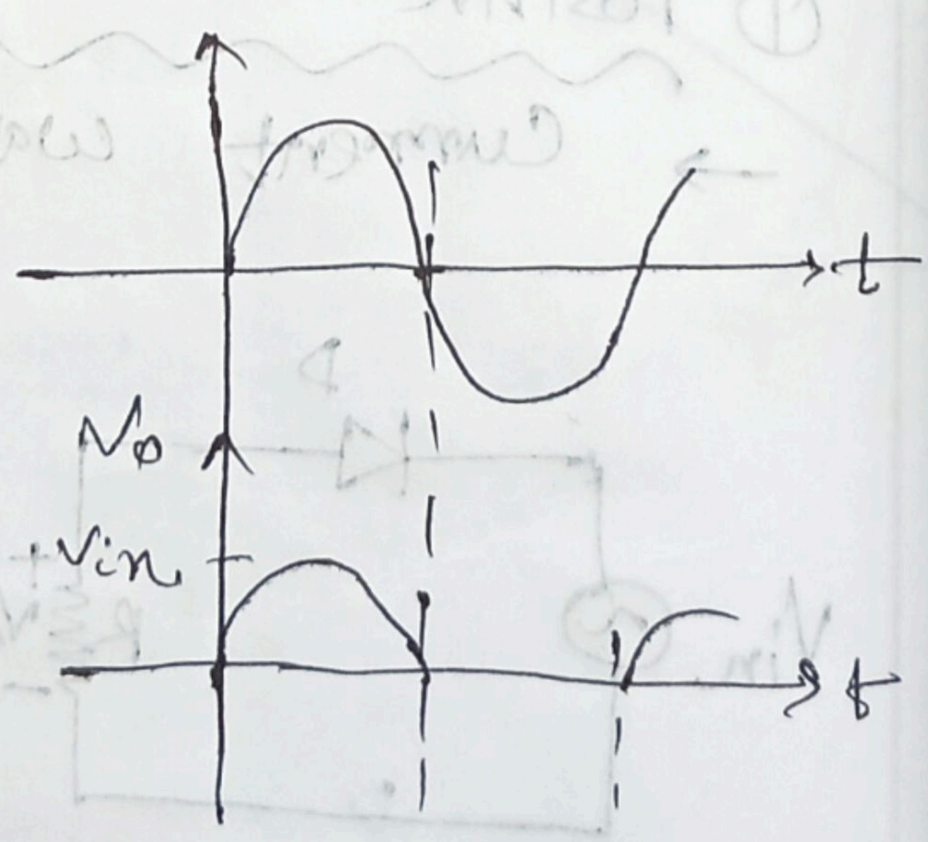
(-)ve,

D = off,

$V_o = -V_{in}$

✱ Negative shunt clipper



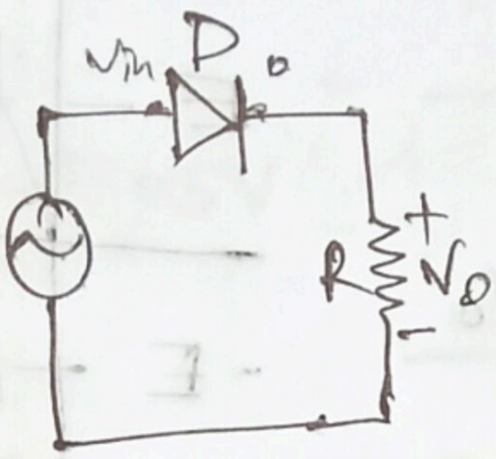


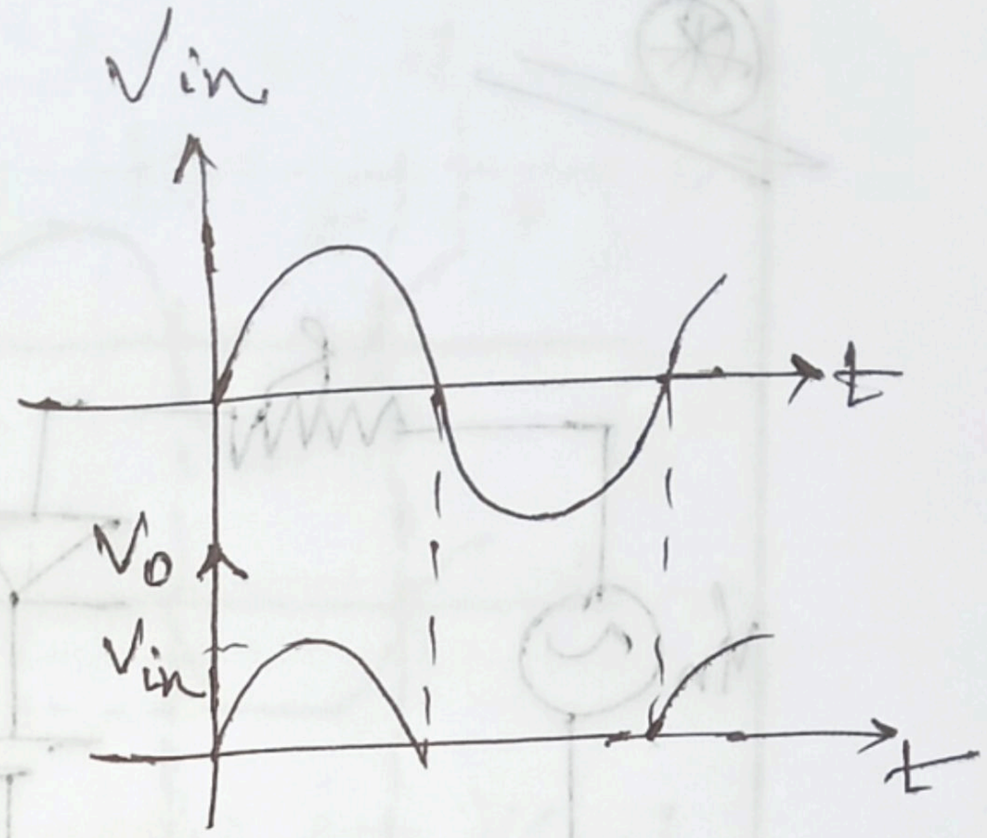
~~(-)ve~~

→ used ~~to~~ clip voltage waveform

⊛ Negative series clipper







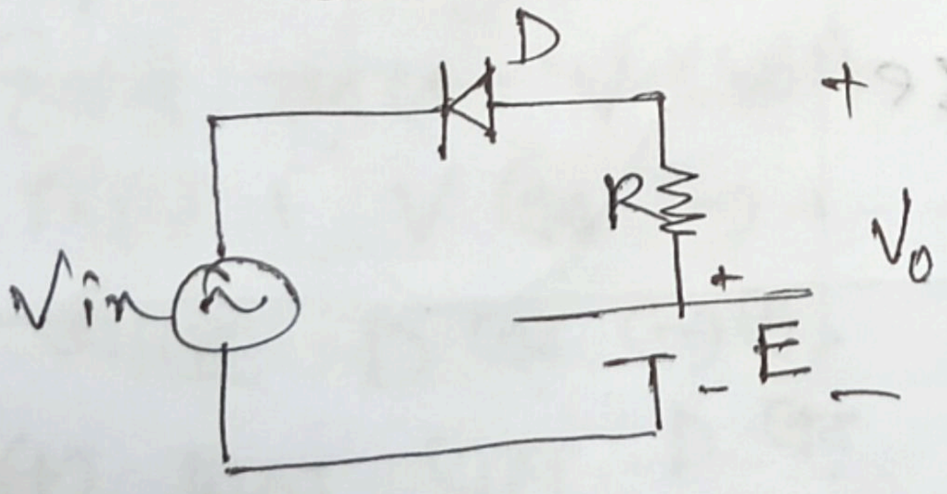
(+)ve,
D = on,
$V_o = V_{in}$

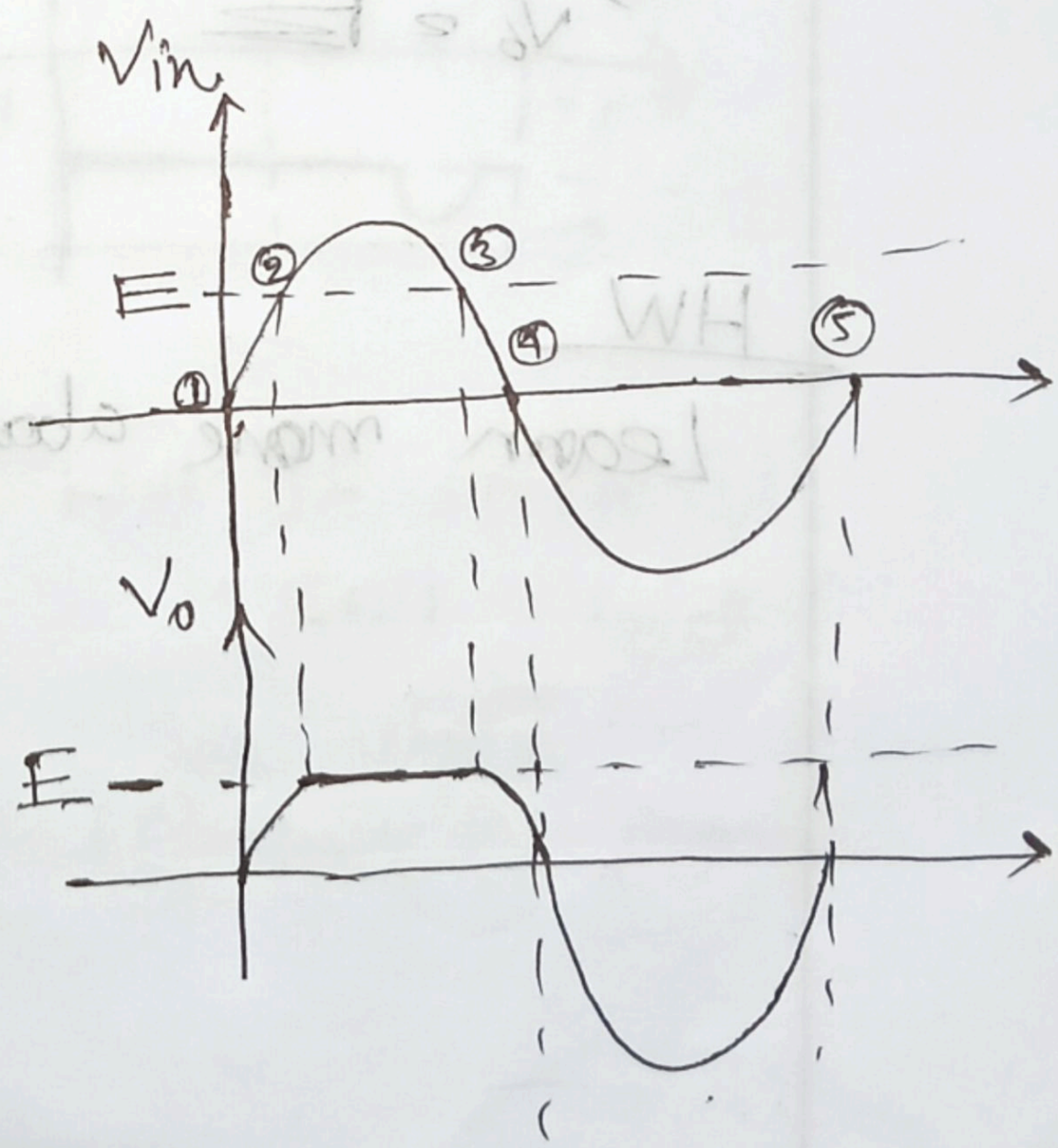
(-)ve,
D = off
$V_o = 0$

→ used to clip current waveform

⊞ Biased clipper

① positive Biased

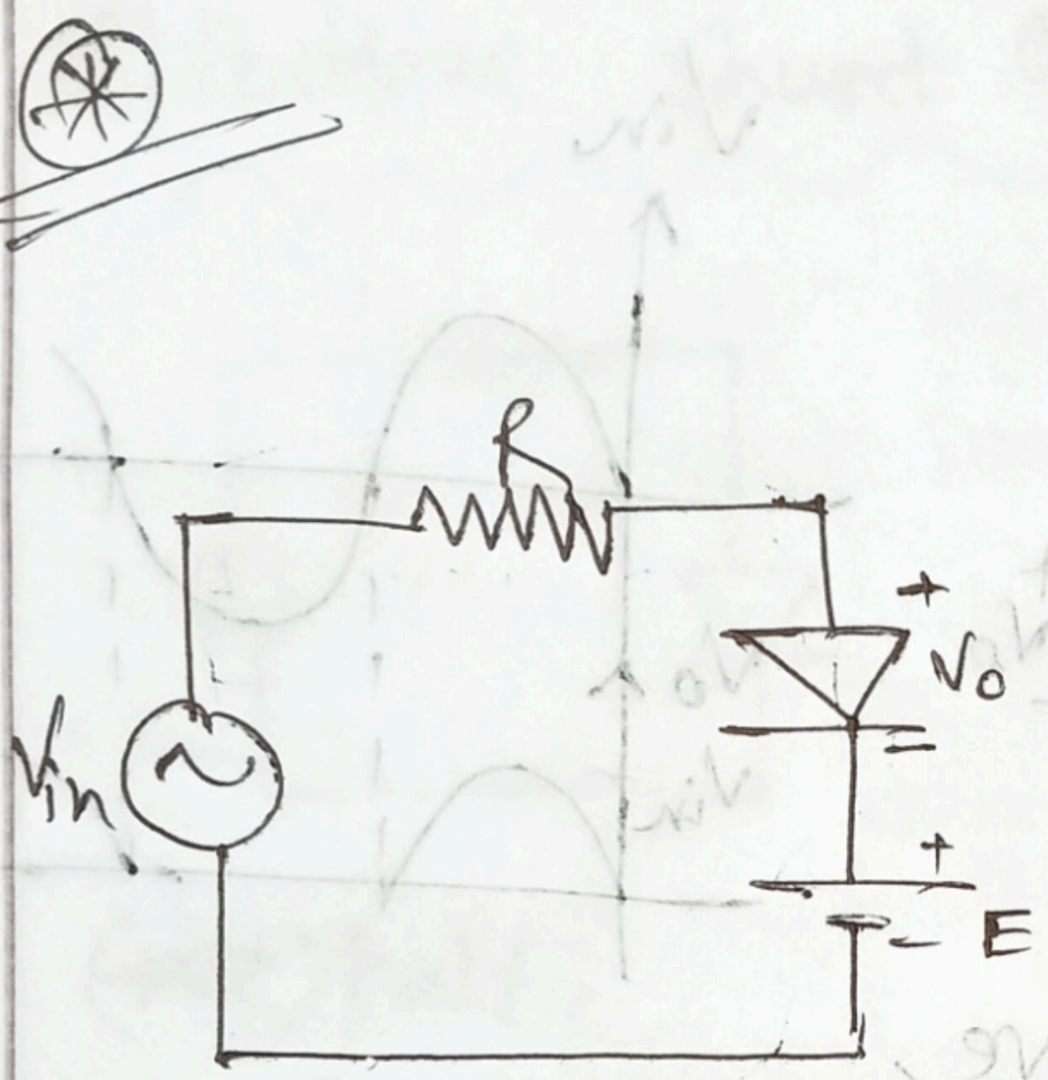


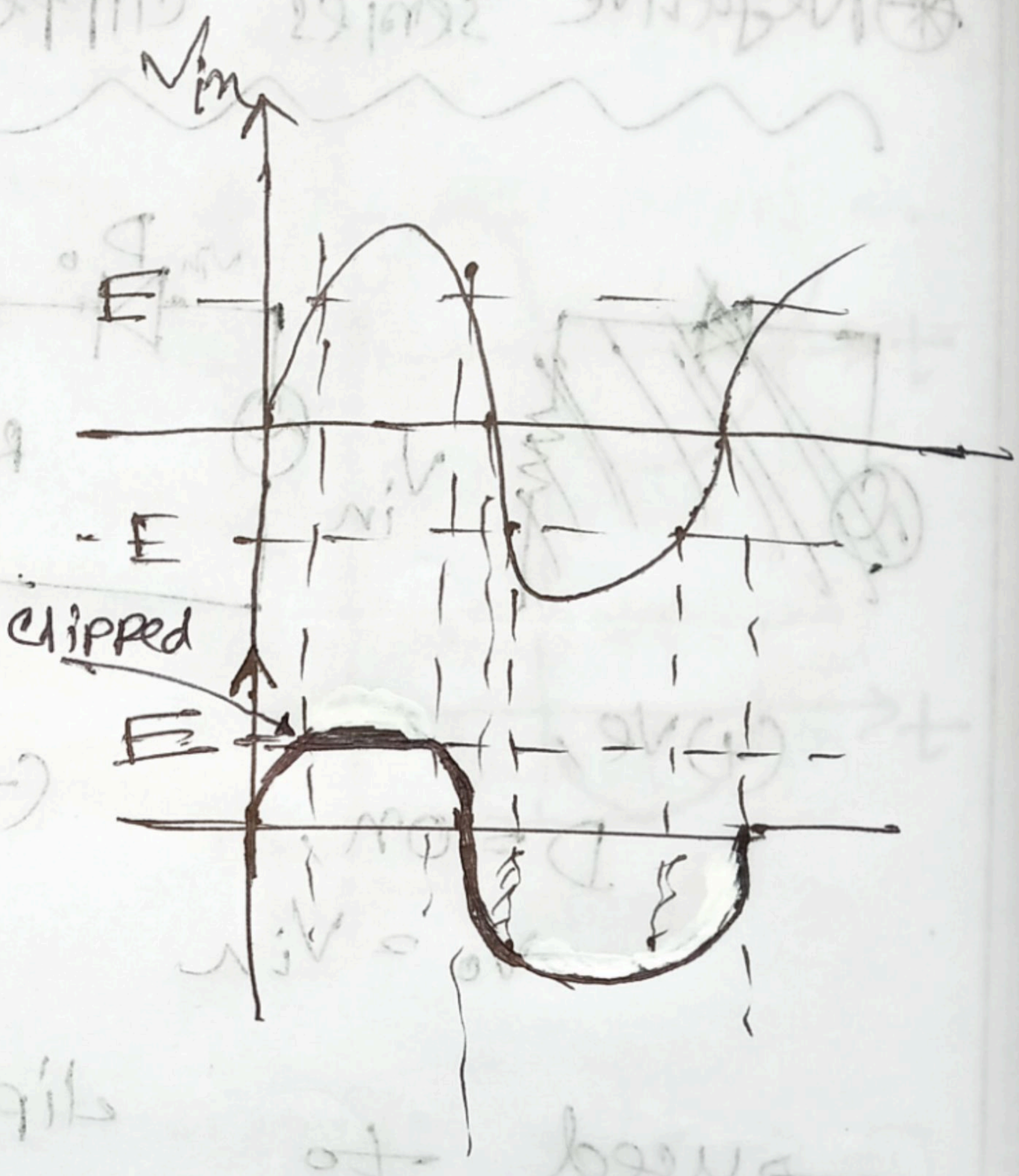


(+)ve half,

$V_{in} > E$ [② → ③]
↳ $V_o = E$

$V_{in} < E$
$V_o = V_{in}$





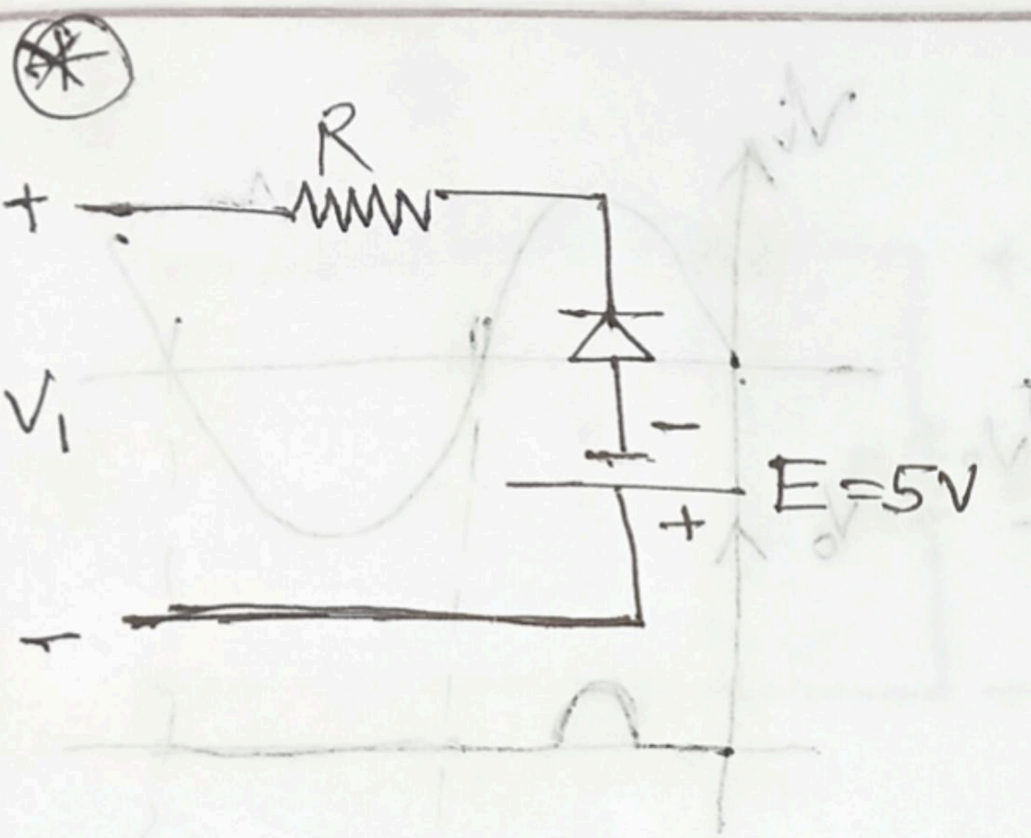
$E < V_{in}$

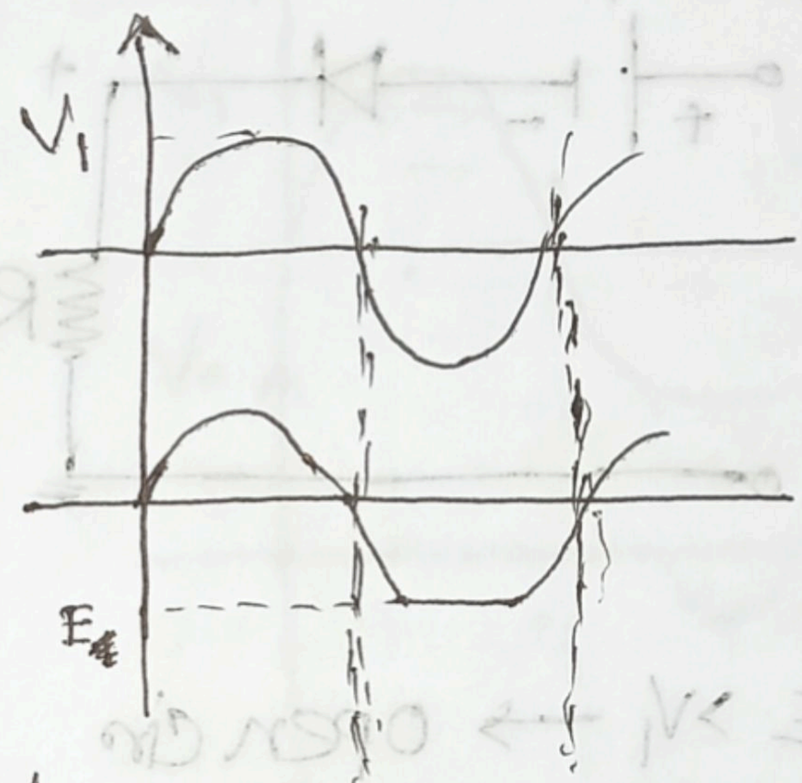
$V_o = E$

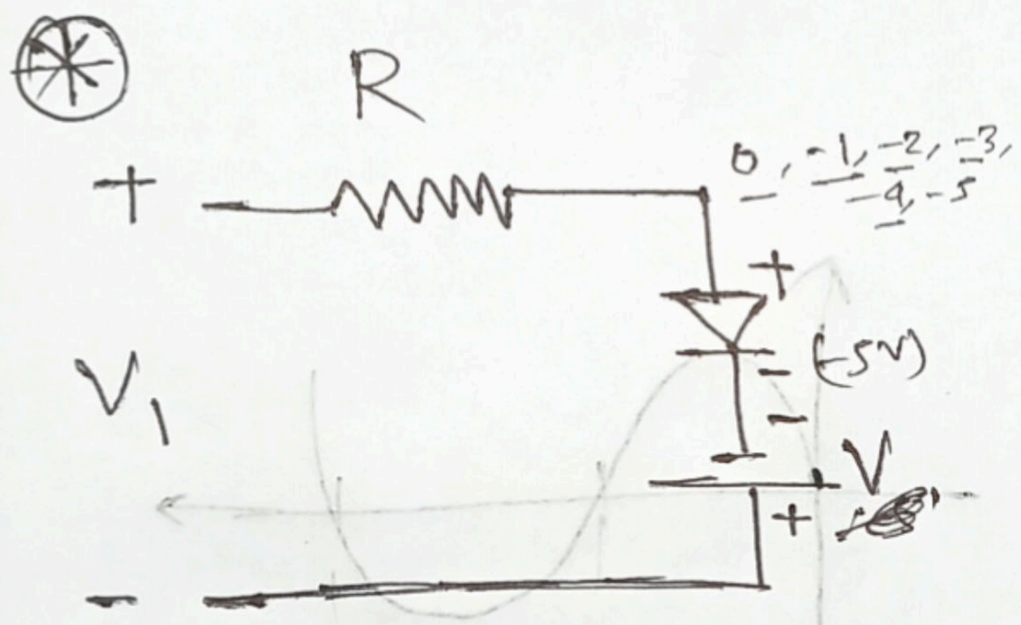
HW

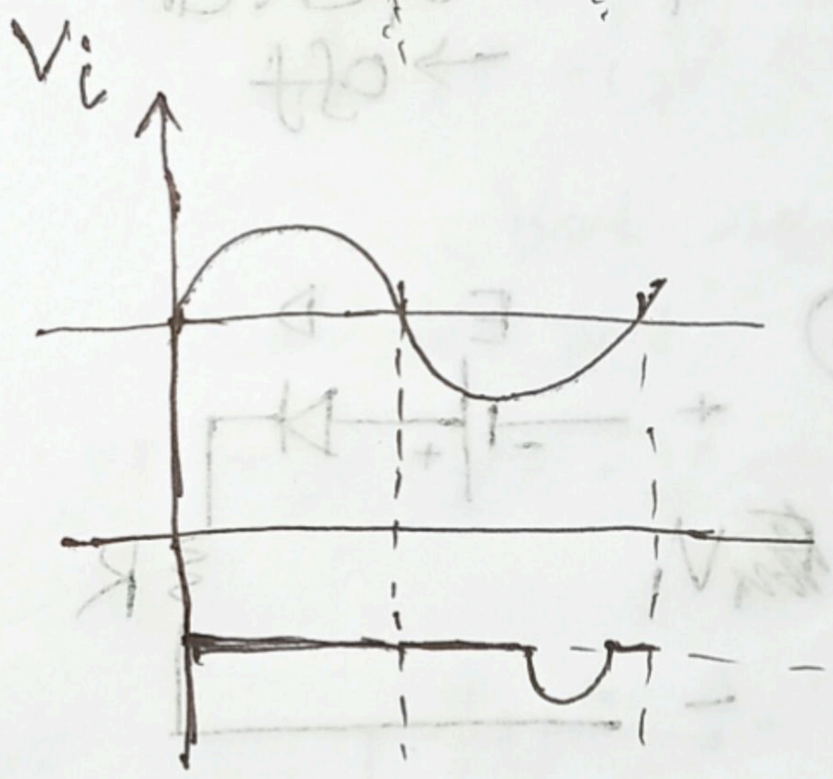
Learn more about these

02.11.26



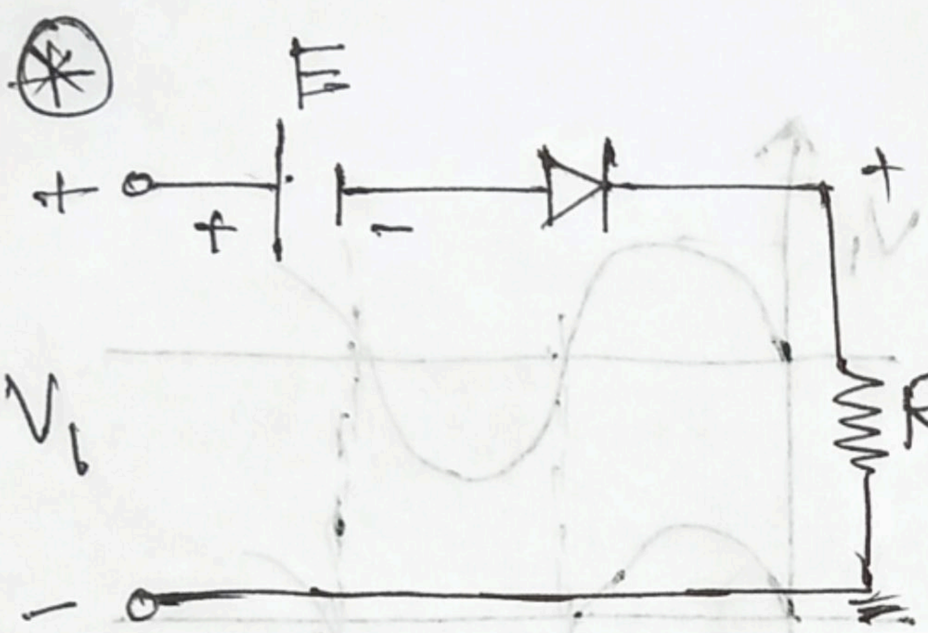


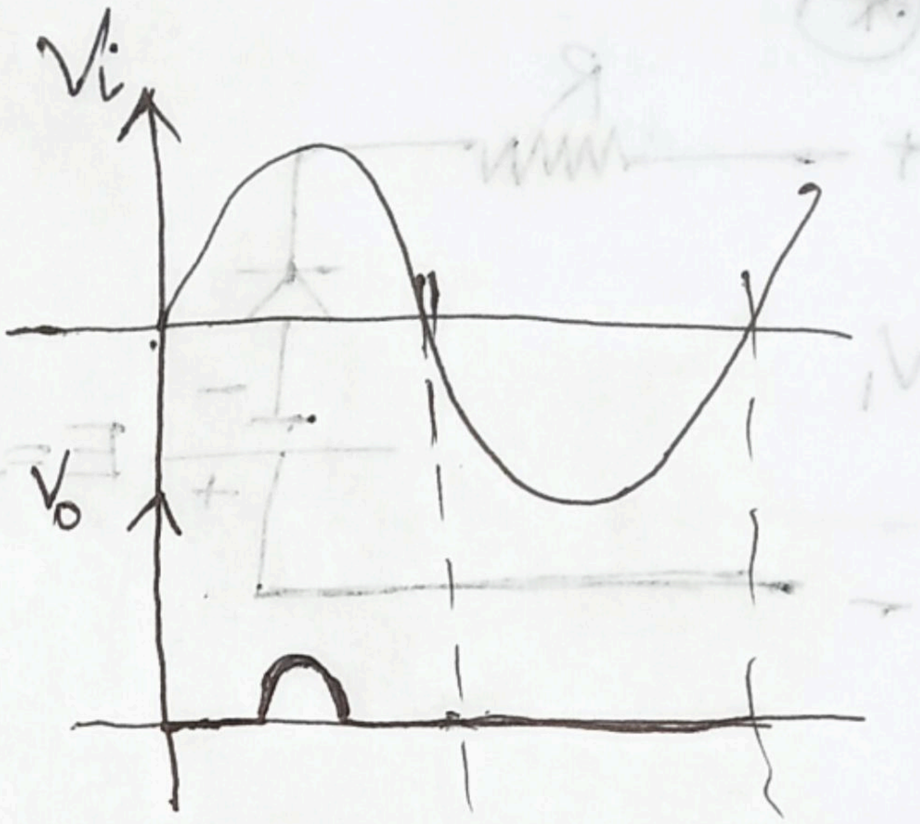




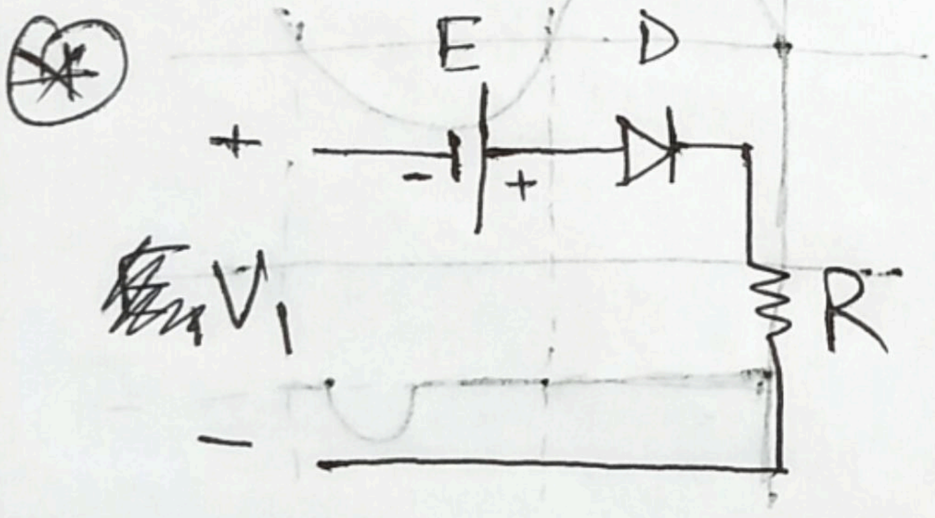
input ছাড়াও current তৈরি করবে 'V' volt দিয়ে। V এর (−) আছে D এর (−) এ। (+) ঘুরে এসে D এর (+) এর সাথে বসে current generate করবে

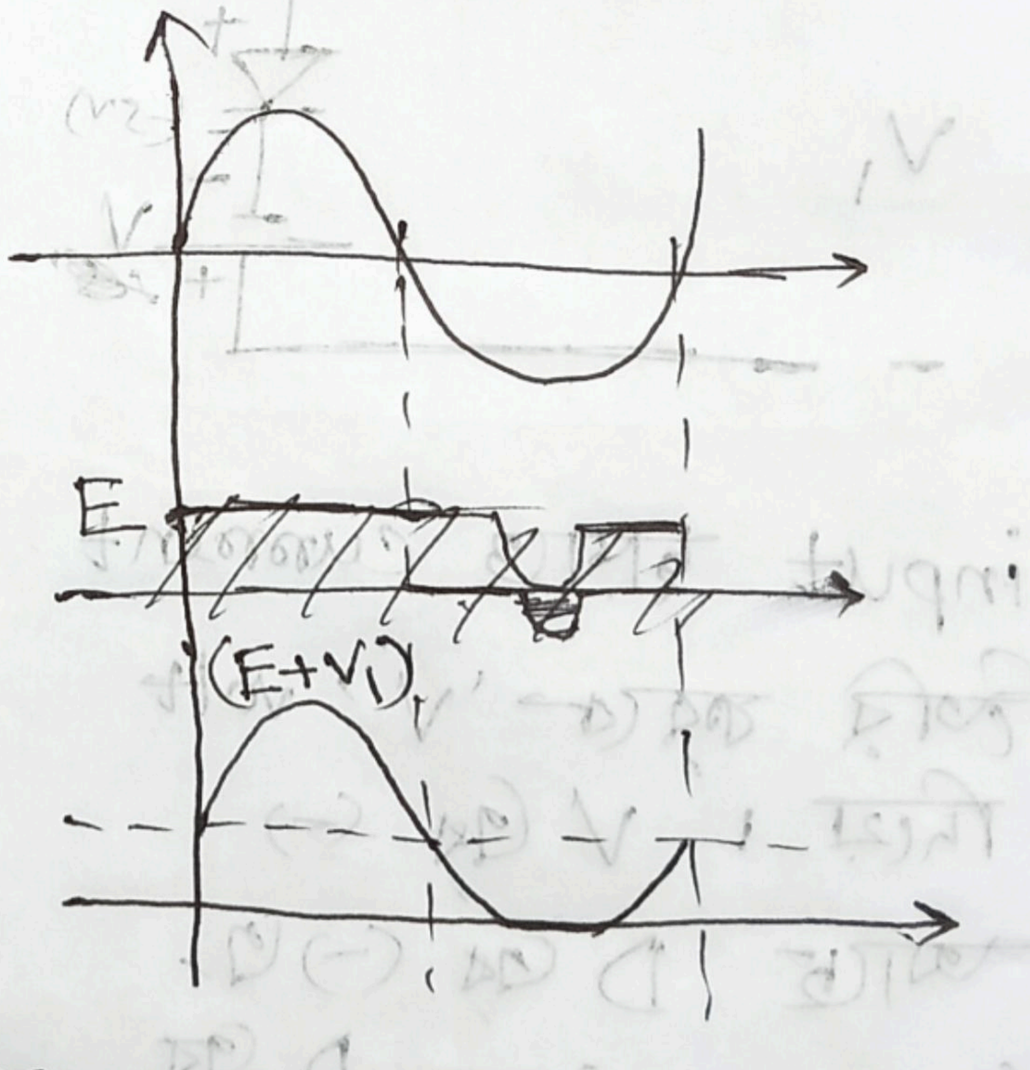
⟶ Short Circuit





E > V_i → open cir
→ off

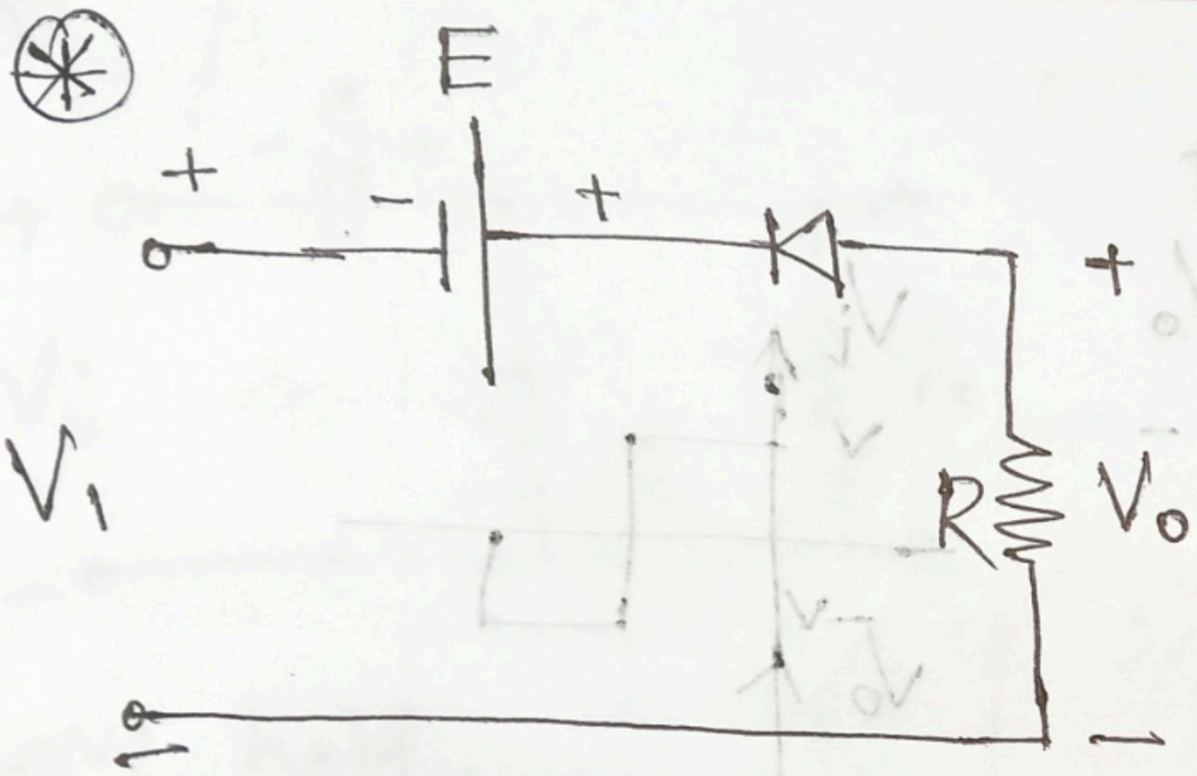


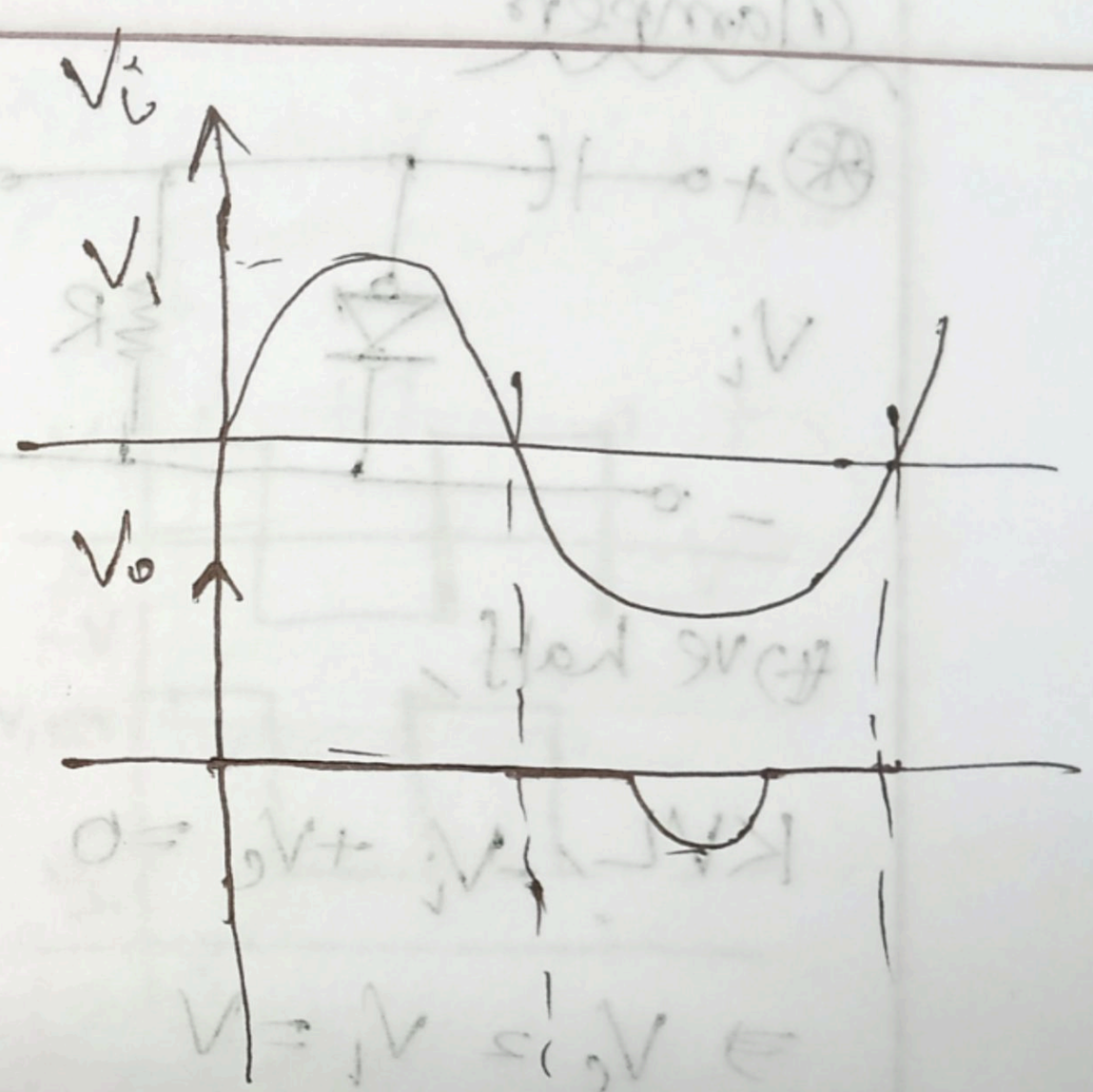


(i) input না দিলে চলে কিনা?

→ চলে।

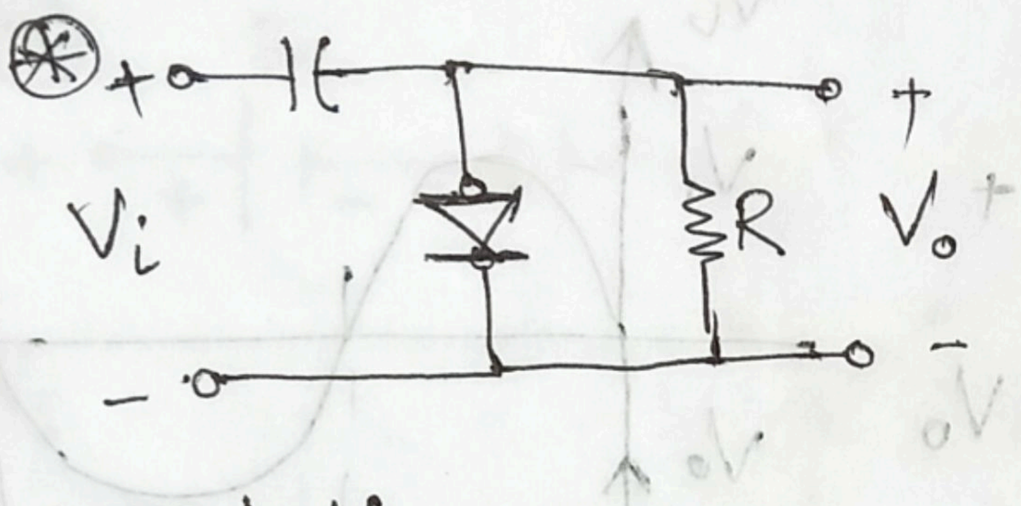
→ 1st cycle এ E এর সমান

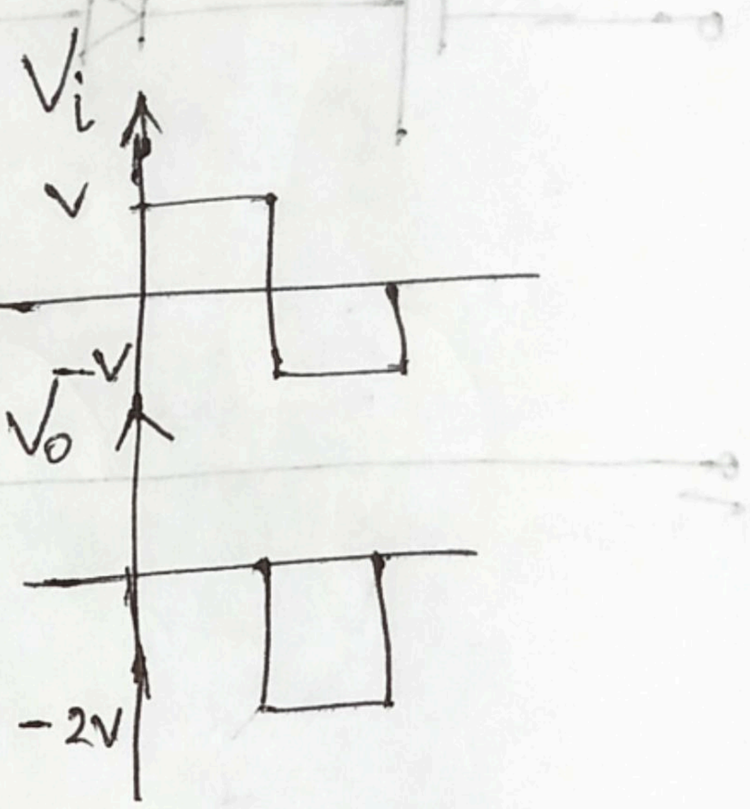

E
+
−
+
+
V1
R
Vo
−


Vi
V1
Vo

03.11.2026

Clamper





+ve half,

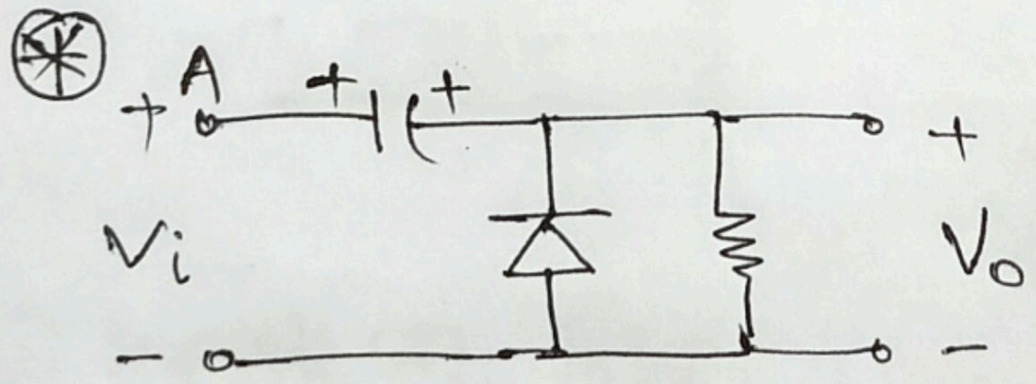
KVL, $-V_i + V_c = 0$

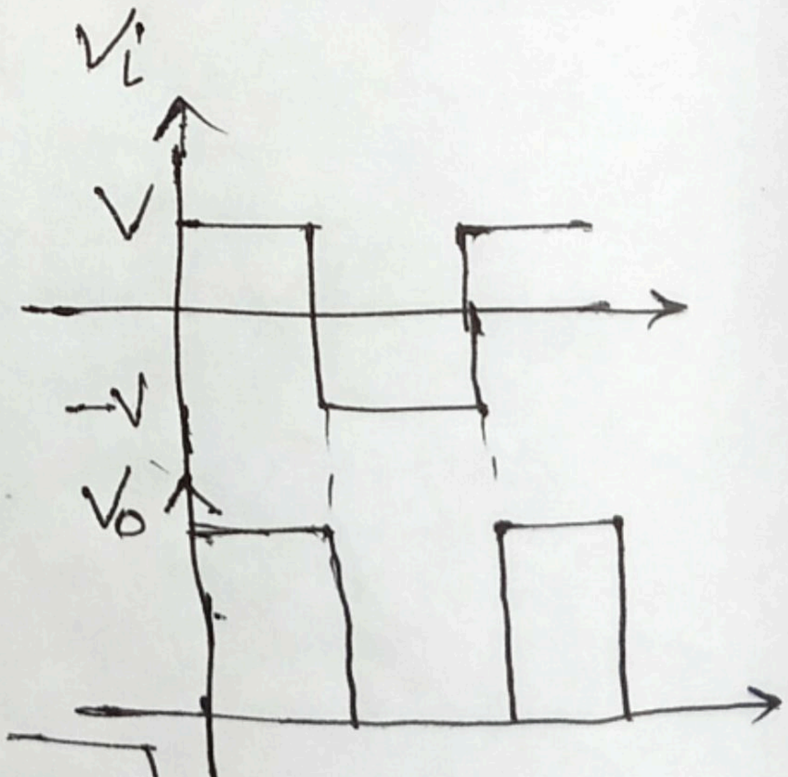
$\Rightarrow V_c = V_i = V$

−ve half,

$+V_i + V_c + V_o = 0$

$\Rightarrow V_o = -V_i - V_c = -2V$





+ve half,

KVL,

$+V_i - V_c = 0$ [এখানে A প্রান্তের চার্জ 3 capacitor এর দুই প্রান্তে same charge]

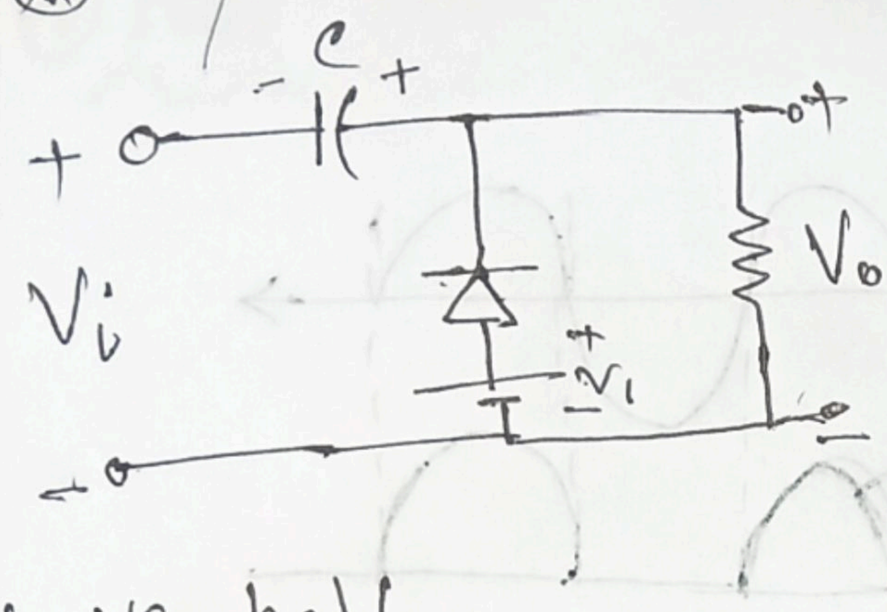
$V_c = V_i = V$

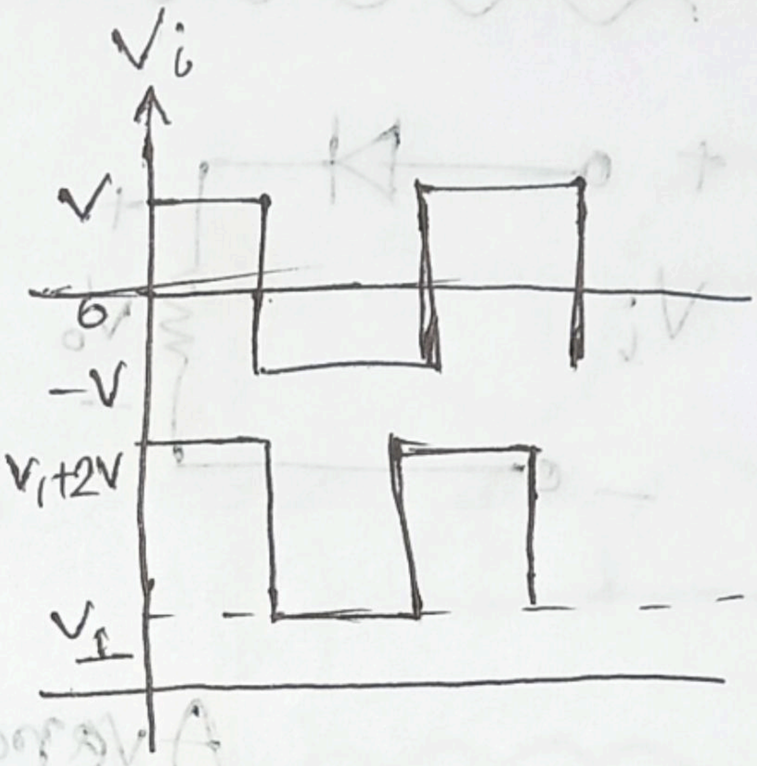
(+)ve half,

$-V_i - V_c + V_o = 0$ [Diode cancel]

$\Rightarrow V_o = V_i + V_c$

$= 2V$





(+)ve half,

$$-V_i + V_c + V_o = 0$$

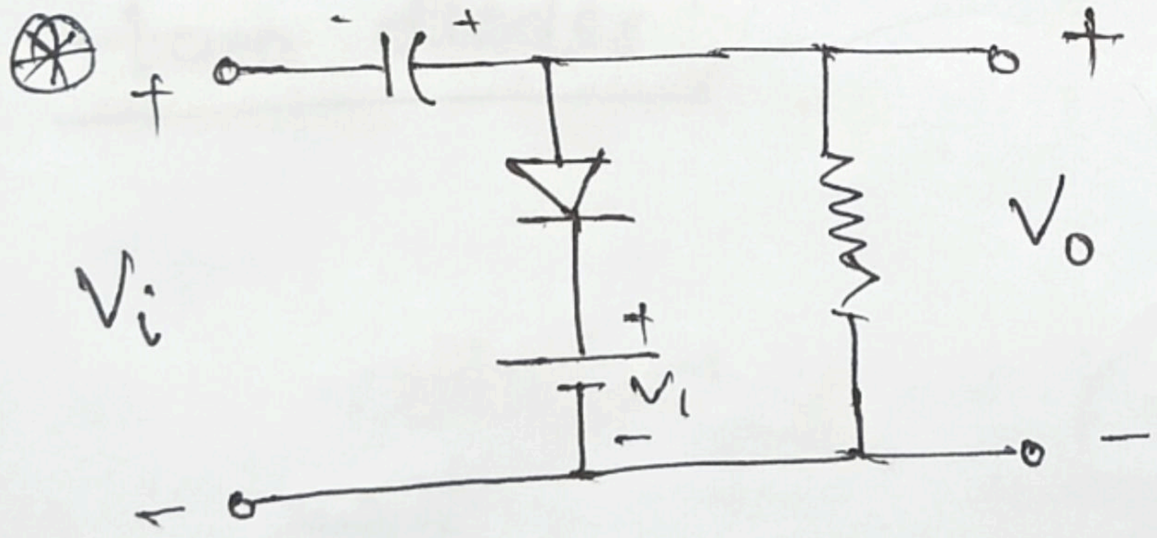
$$\Rightarrow V_o = V_i + V_c$$

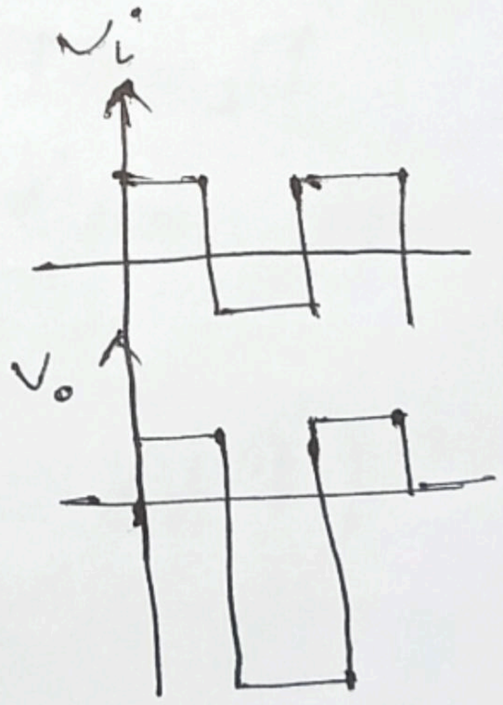
$$= V_1 + 2V$$

(−)ve half,

$$+V_i + V_c + V_1 = 0$$

$$\Rightarrow V_c = V_i + V_1 = 2V$$





(+)ve half,

$$+V_i + V_c + V_1 = 0$$

$$\Rightarrow V_c = -V_i + V_1 = -V - V + V_1 = -2V + V_1$$

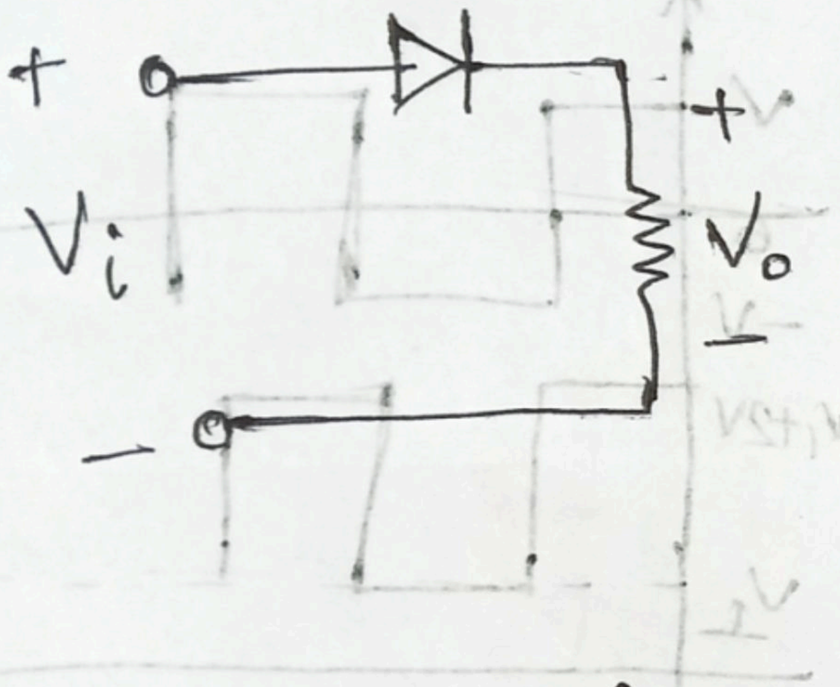
(+)ve half,

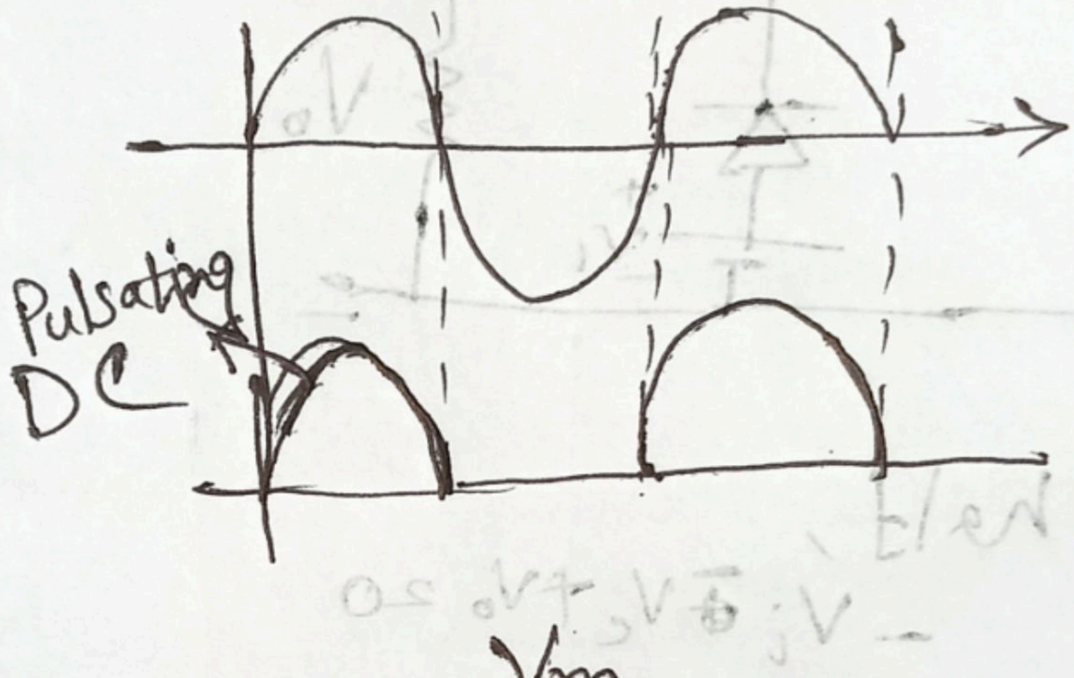
$$-V_i + V_c + V_1 = 0$$

$$\Rightarrow V_c = V - V_i +$$

$V_o = V_1$

Half Wave Rectifier



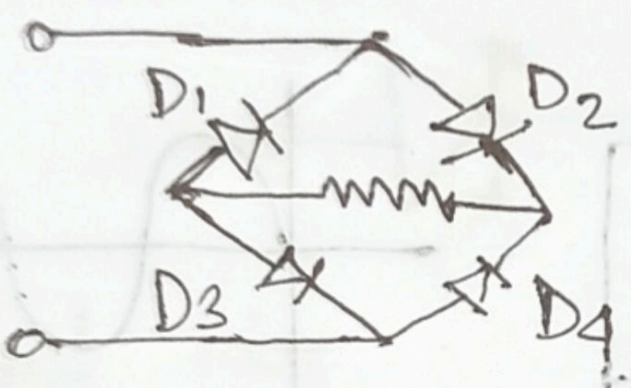


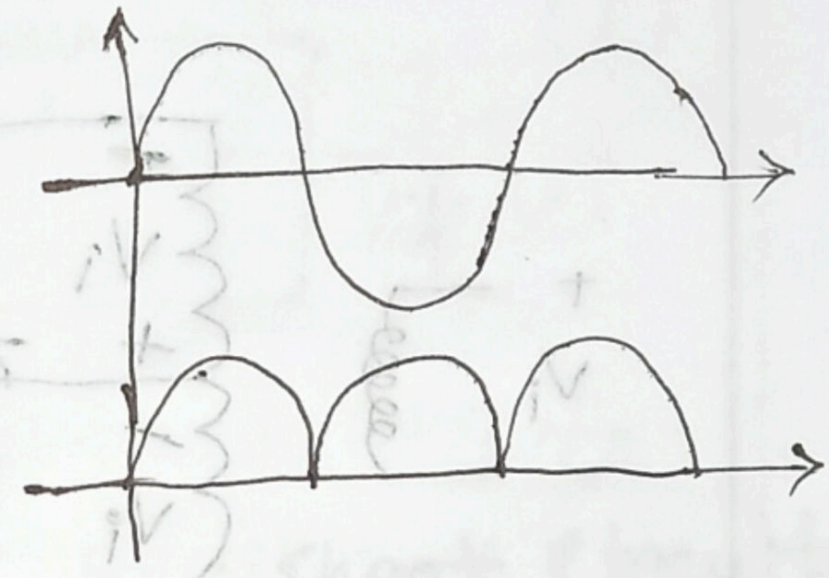
$$\text{Average, } V_{dc} = \frac{V_m}{\pi}$$

$$= 0.318\, V_m$$

04.11.26

✳ Full Wave Bridge Rectifier





(+)ve half,

D_2, D_3 = ON

PIV = V_m

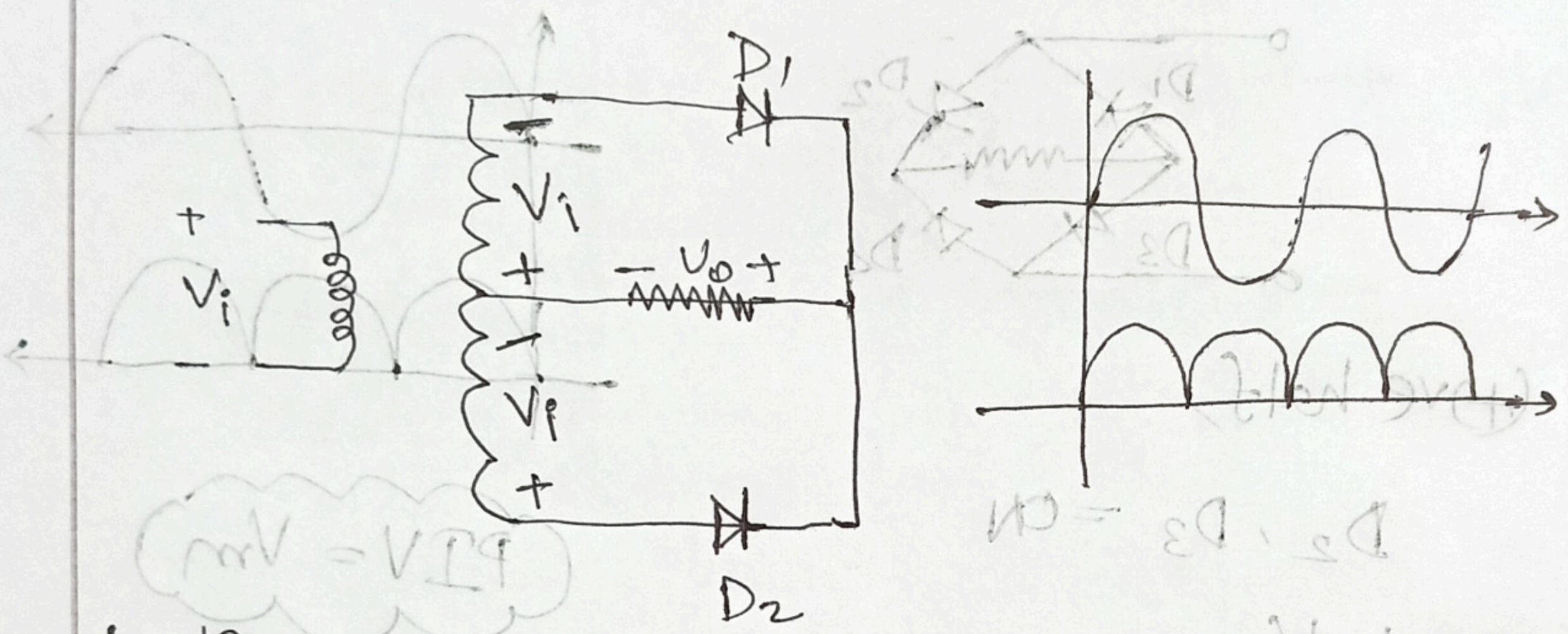
(-)ve half,

D_1, D_4 = ON

⇒ How to make a full wave rectifier using two diodes:

25.11.20

Center tapped transformer



(+)ve,

D_1 = ON,

$V_o = V_i$

D_2 = OFF,

V_o = 0

(-)ve,

D_1 = OFF

D_2 = ON;

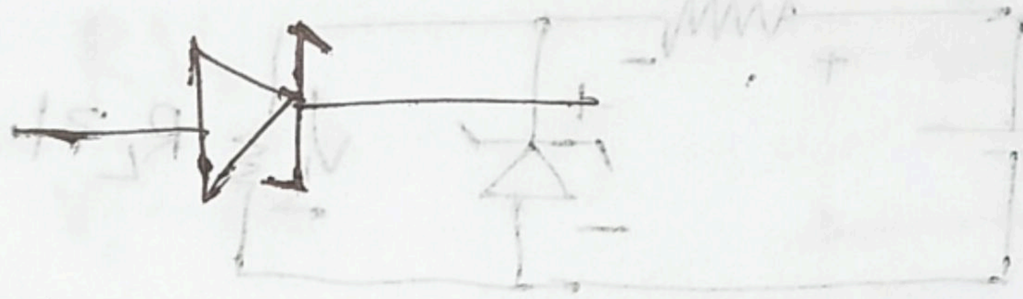
$V_o = V_i$

PIV = $2V_m$

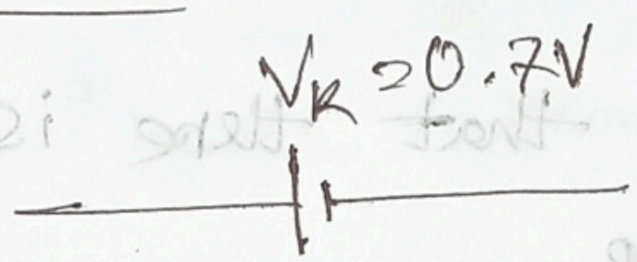
⇒ Better than bridge

Peak Inverse Voltage

Zener Diode

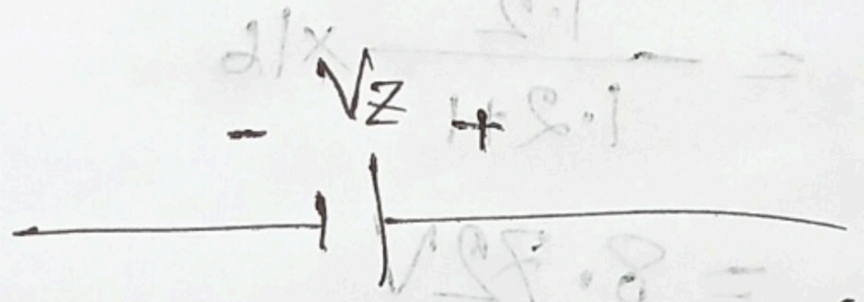


Forward bias



short circuit after threshold voltage (V_K)

Reverse bias

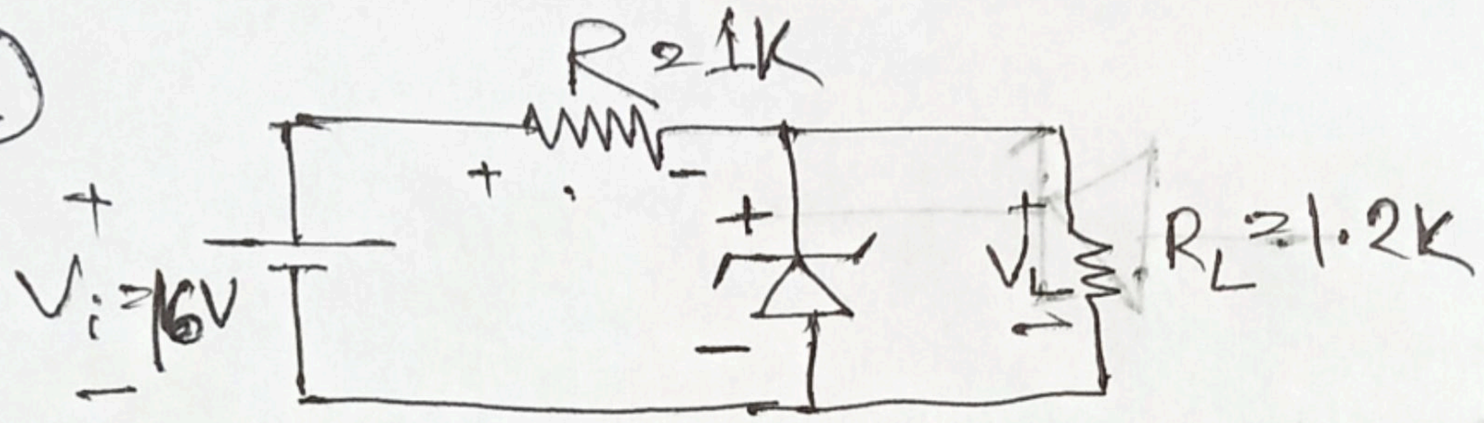


After V_Z Voltage it will work

if the $V_Z = 10V$; it will always convey that exact voltage. even if we give 50V, it will only convey 10V; but if we give 9V, it will work as an open circuit.

(i) V_i & R fixed

V_L, V_R, I_Z, P_Z

① 

step - 1

considering that there is no diode

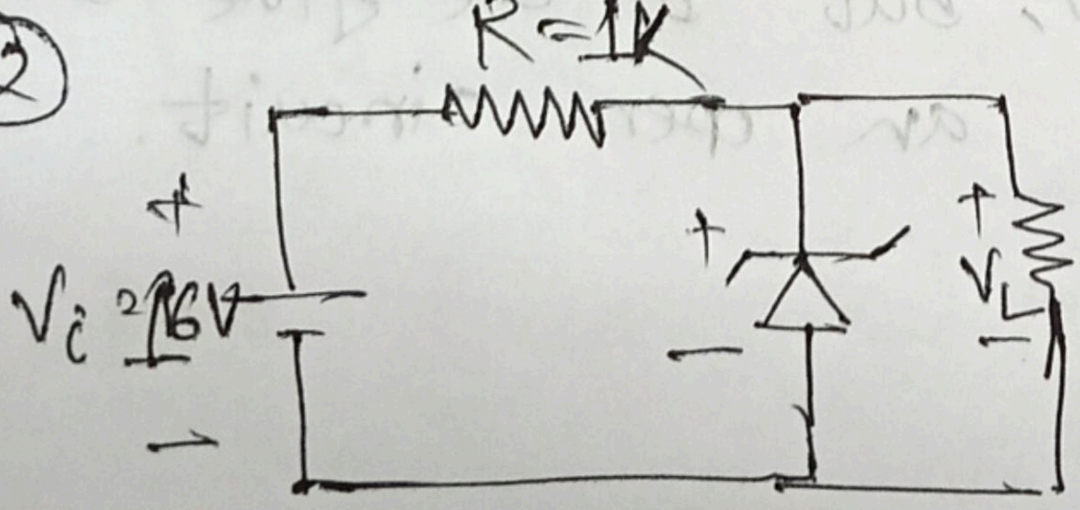
$$V_L = \frac{R_L}{R+R_L} \times V_i$$

$$= \frac{1.2}{1.2+1} \times 16$$

$$= 8.72V$$

Here, $V_L = V_Z$ as they are in paralleⅼ

$V_Z = V_L < 10V$, So zener diode would be off

② 

$$V_L = \frac{R_L}{R+R_L} \times V_i$$

$$= 12V$$

$V_Z = 12V > 10V$, Diode ON

ON হলেই V_L এর মান থাকে 10V

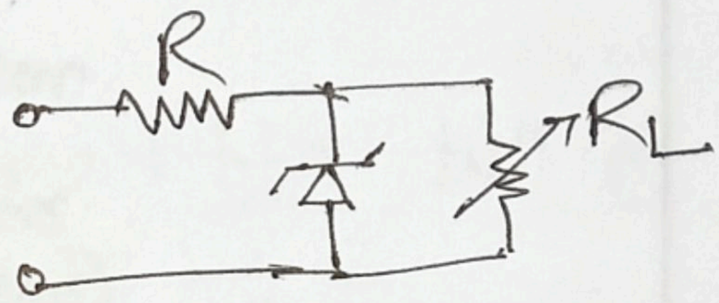
$V_L = 10V$

$V_R = (16-10)V$

$= 6V$

(ii) V_i = Fixed & R_L = Variable

$$R_{L\min} = \frac{R V_Z}{V_i - V_Z}$$



$$I_{L\,max} = \frac{V_Z}{R_L} = \frac{V_Z}{R_{L\min}}$$

$$I_{L\min} = I_R - I_{Z\,m}$$

$$V_R = V_i - V_Z$$

$$R_{L\max} = \frac{V_Z}{I_{L\min}}$$

$$I_R = \frac{V_R}{R} \qquad I_Z = I_R - I_L$$

Ex - 2.27

(iii) V_i = Variable & R_L = Fixed

Ex - 2.28